



TCC6131 CLOUD COMPUTING

PROJECT

Term 2420

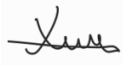
No.	Student ID	Student Name	Majoring	Signature
1.	1211103698	UMMI SYAHIRAH BINTI MUHAMMAD ROZAIDEE @ ROZI	BIO	
2.	1211110723	U BOON XUAN	BIO	
3.	1221301543	NUR FARHAH WAJIAH BINTI MUHAMAD ZAID	BIO	
4.	1221302732	HANI HAMIZAH BINTI HAMZAN	ST	

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Acknowledgement

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Furthermore, I also extend my sincere appreciation to Mr. Roy Chang Kwang Yang and Dr. Ramadhan Mazen Farid Ebrahim, our lab instructors, for their hands-on assistance and practical explanation during lab sessions. Your patience and willingness to clarify doubts have been invaluable in strengthening our understanding of cloud computing technologies and have the opportunity experience AWS provided service in this cloud computing subject.

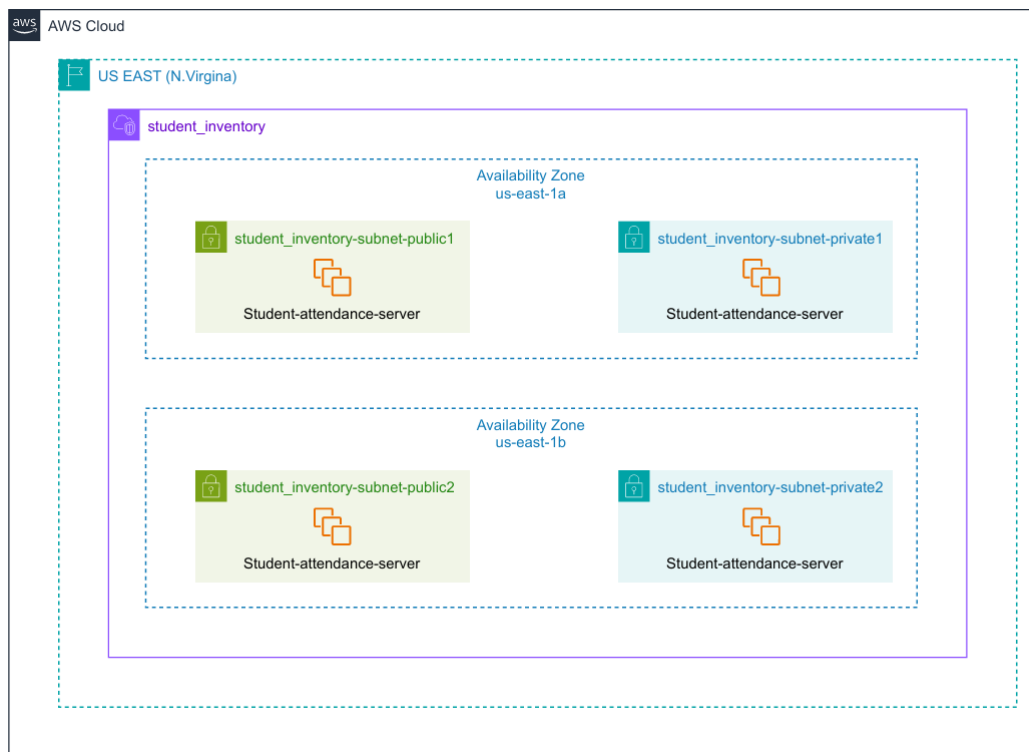
Lastly, I would like to acknowledge the efforts of my assignment and project teammates, for hard work, collaboration, and ideas were essential in completing this project. Working with such a committed and enthusiastic group made this experience truly rewarding.

1.0 Phase 1

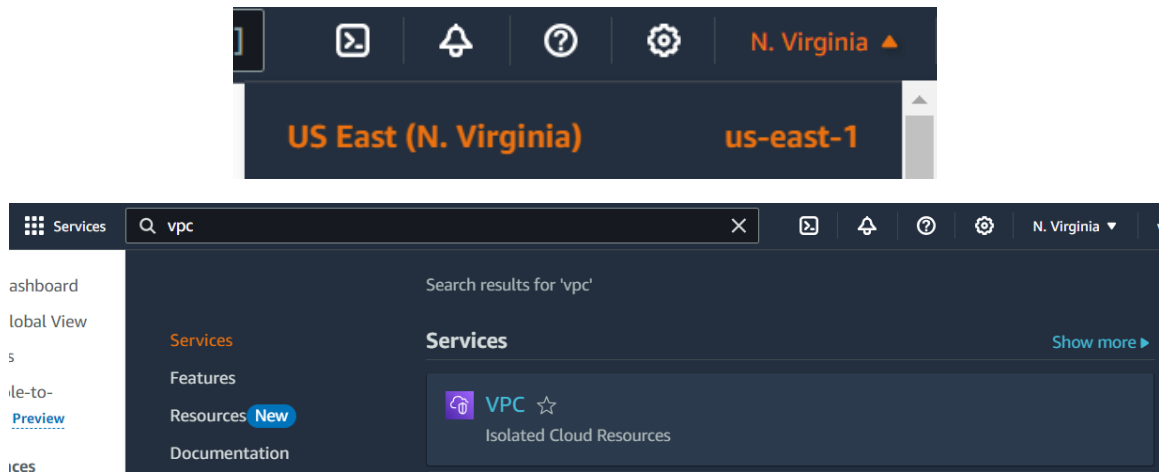
1.1 Introduction of Student Attendance System

In this project, we build an online student attendance inventory from the web-based application designed specifically for teachers, lecturers, or tutors in educational institutions. This student attendance inventory has features and functionalities that can easily record and track student daily attendance. This system does not require a sign in or log-in section to access its primary features and functionalities. Instructors can directly add new classes, edit classes, and delete classes that they teach in the class management section. Additionally, instructors can also manage the student data, such as adding a new student, editing a student, and deleting a student from the inventory in the student management section. Users may select the class and date before continuing to the attendance form. In the attendance form, users can view the list of student's data with a “**check/uncheck attendance status**” to update student's attendance.

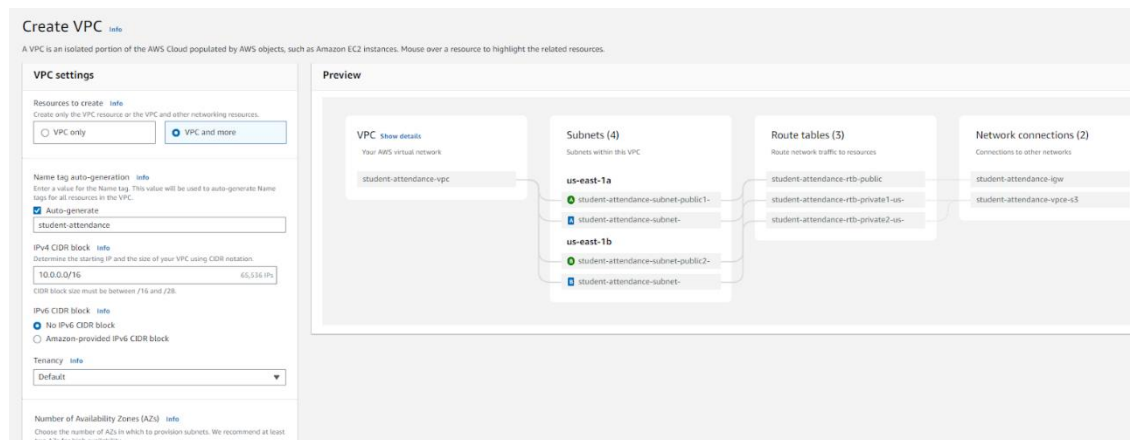
1.2 Layout diagram of the network



1.3 Implementation steps



1. Make sure N.Virginia was chosen as a selected region. Search for VPC at the search bar and choose “Create VPC” to create a VPC.



2. In the VPC settings, choose “ VPC and more” and put the name tag as a “student-attendance”.

IPv6 CIDR block

Info

☒ No IPv6 CIDR block
 ☐ Amazon-provided IPv6 CIDR block

Tenancy

Info

Default

Number of Availability Zones (AZs)

Info

Choose the number of AZs in which to provision subnets. We recommend at least two AZs for high availability.

[Customize AZs](#)

Number of public subnets

Info

The number of public subnets to add to your VPC. Use public subnets for web applications that need to be publicly accessible over the internet.

Number of private subnets

Info

The number of private subnets to add to your VPC. Use private subnets to secure backend resources that don't need public access.

[Customize subnets CIDR blocks](#)

NAT gateways (\$)

Info

Choose the number of Availability Zones (AZs) in which to create NAT gateways. Note that there is a charge for each NAT gateway.

VPC endpoints

Info

Endpoints can help reduce NAT gateway charges and improve security by accessing S3 directly from the VPC. By default, full access policy is used. You can customize this policy at any time.

DNS options

Info

☒ Enable DNS hostnames
 ☒ Enable DNS resolution

[Additional tags](#)

Cancel

Create VPC

3. The screenshot above shows the configuration settings and click “Create VPC”.

Success

Details

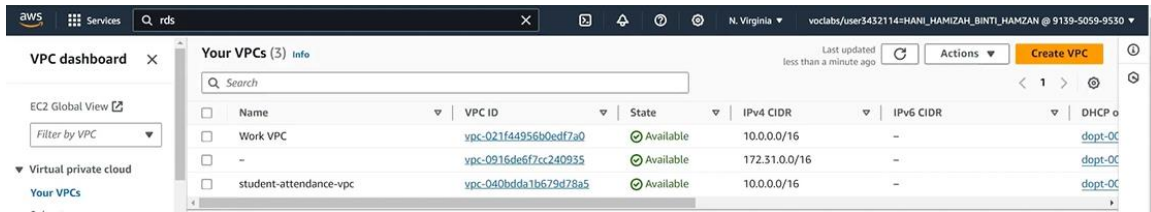
- ✔ Create VPC: vpc-040bdda1b679d78a5
- ✔ Enable DNS hostnames
- ✔ Enable DNS resolution
- ✔ Verifying VPC creation: vpc-040bdda1b679d78a5
- ✔ Create S3 endpoint: vpce-02aea428285031fab
- ✔ Create subnet: subnet-03c9e8ad2427f036e
- ✔ Create subnet: subnet-00bf0776133562406
- ✔ Create subnet: subnet-0d4187c8954ebdfa1
- ✔ Create subnet: subnet-028261de73ae1c9da
- ✔ Create internet gateway: igw-0b95cd24c3f289892
- ✔ Attach internet gateway to the VPC
- ✔ Create route table: rtb-07b99267488b81ca7
- ✔ Create route
- ✔ Associate route table
- ✔ Associate route table
- ✔ Create route table: rtb-013795c5e40261c8e
- ✔ Associate route table
- ✔ Create route table: rtb-0b01309d000997501
- ✔ Associate route table
- ✔ Verifying route table creation
- ✔ Associate S3 endpoint with private subnet route tables: vpce-02aea428285031fab

View VPC

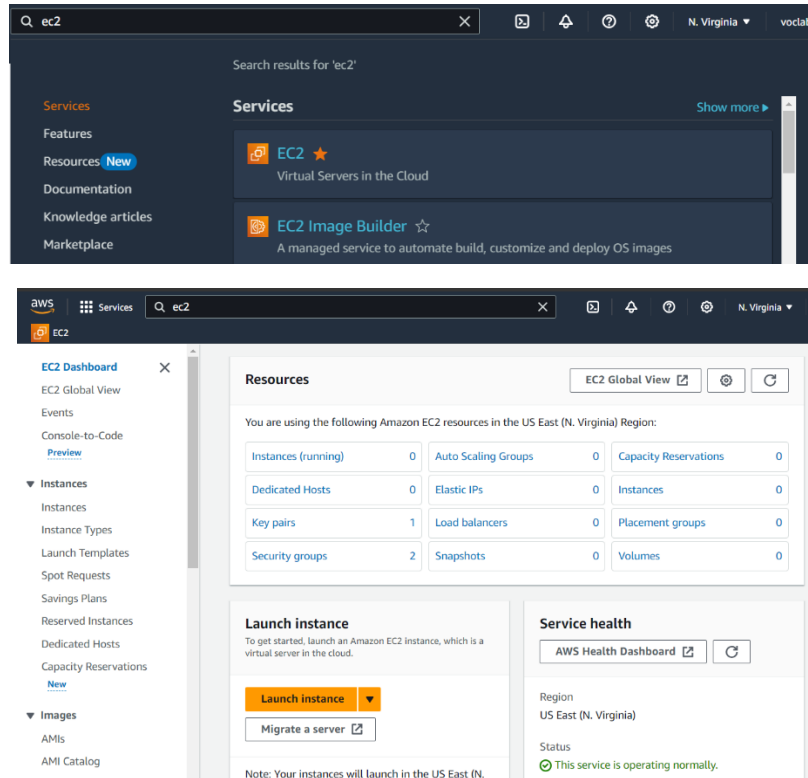
CloudShell

Feedback

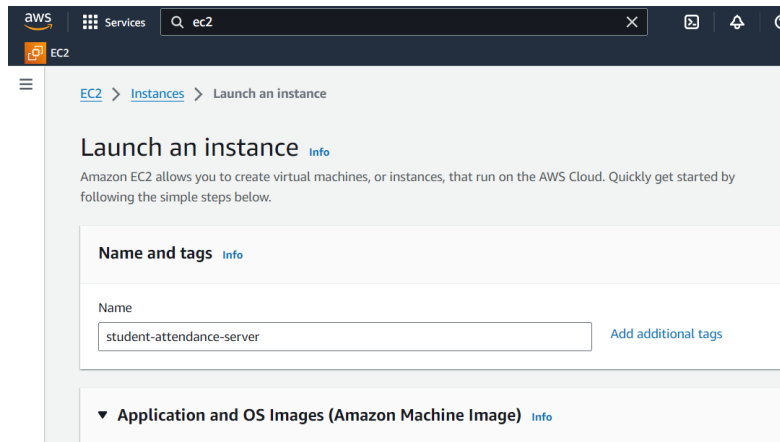
© 2024, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences



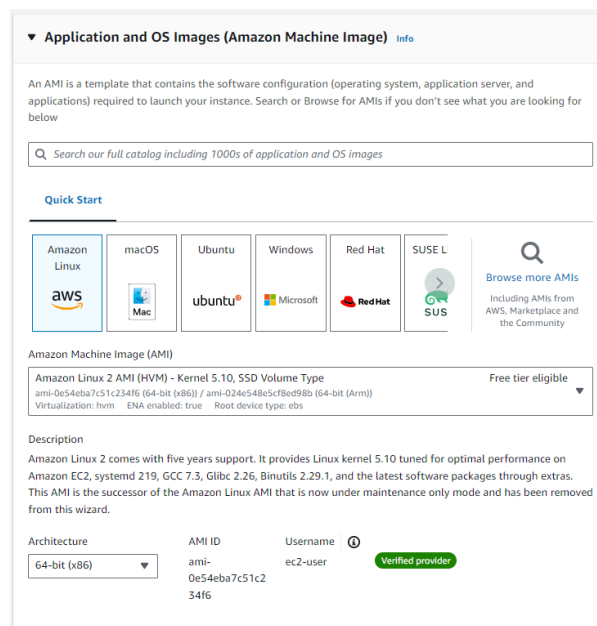
4. The created VPC has shown in the VPC dashboard.



5. Next step, search EC2 in the search bar to create an instance and click the “Launch instance” to create a new instance.



6. Name your instance as a “student-attendance-server”.



7. In the **Application and OS Images (Amazon Machine Image)**, choose Amazon Linux and Amazon Linux 2 AMI (HVM) - Kernel 5.10, SSD Volume Type in the Amazon Machine Image (AMI).

Create key pair

Key pair name
Key pairs allow you to connect to your instance securely.

stud-att-key

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

Cancel Create key pair

8. In Key pair (login), create a new key pair. Fill up the key pair name as a “stud-att-key” and setup the rest of the settings as in the screenshot above.

▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

stud-att-key

↻ Create new key pair

9. After creating a new key pair, select the key pair that we created just now.

▼ Network settings Info

VPC - required Info

vpc-040bdda1b679d78a5 (student-attendance-vpc)

Subnet Info

subnet-03c9e8ad2427f036e student-attendance-subnet-public1-us-east-1a

VPC: vpc-040bdda1b679d78a5 Owner: 9118950599530
Availability Zone: us-east-1a Zone type: Availability Zone
IP addresses available: 4091 CIDR: 10.0.0.0/20

Auto-assign public IP Info

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

Security group name - required

server-security-group

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and ., /, !, @, [] + = & ; ' *

Description - required Info

security group

Inbound Security Group Rules

▼ Security group rule 1 (TCP; 22, 0.0.0.0/0) Remove

Type: [Info](#) Protocol: [Info](#) Port range: [Info](#)

Source type: [Info](#) Source: [Info](#) Description - optional: [Info](#)

▼ Security group rule 2 (TCP; 80, 0.0.0.0/0) Remove

Type: [Info](#) Protocol: [Info](#) Port range: [Info](#)

Source type: [Info](#) Source: [Info](#) Description - optional: [Info](#)

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

10. In Network settings, click “Edit” and change the VPC to “**vpc..... (student-attendance-vpc)**”. Subnet change to **public1**. Auto-assign public IP choose **Enable**. In Firewall (security groups), choose “Create security group” and name the security group (server-security-group). In the security group rule, add a new security rule as “HTTP” and set the source type to “Anywhere”.

Advanced details [Info](#)

Domain join directory: [Info](#) Create new directory

IAM instance profile: [Info](#) Create new IAM profile

Hostname type: [Info](#)

DNS Hostname: [Info](#) ☒ Enable IP name IPv4 (A record) DNS requests ☐ Enable resource-based IPv4 (A record) DNS requests ☐ Enable resource-based IPv6 (AAAA record) DNS requests

Instance auto-recovery: [Info](#)

Shutdown behavior: [Info](#)

Stop - Hibernate behavior: [Info](#)

Summary

Number of instances: [Info](#)

Software Image (AMI): [Info](#) Amazon Linux 2 Kernel 5.10 AMI...read more ami-0e54eb7c31c234f6

Virtual server type (instance type): [Info](#) t2.micro

Firewall (security group): [Info](#) New security group

Storage (volumes): [Info](#) 1 volume(s) - 8 GiB

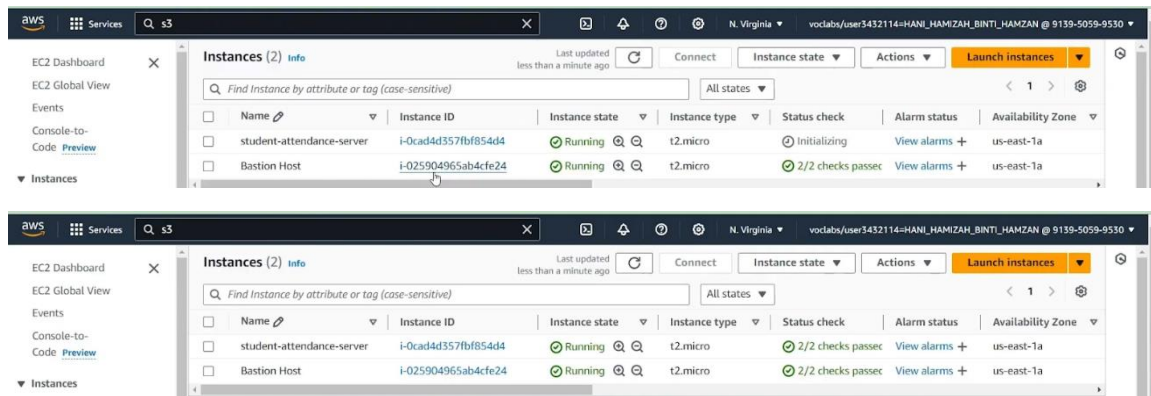
Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance

Cancel Launch instance Review commands

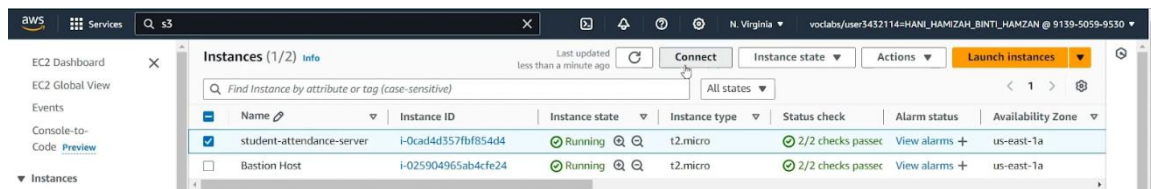
11. In advance details, set the IAM instance profile to “LabInstanceProfile” and shutdown behavior select “stop”.

Success

Successfully initiated launch of instance (i-0cad4d357fbf854d4)



12. After finish setting up an instance, wait the status check of the instance changed to “2/2 checks passed”.



13. Next, go to ‘Connect’.

Connect to instance Info

Connect to your instance i-0cad4d357fbf854d4 (student-attendance-server) using any of these options

[EC2 Instance Connect](#) | [Session Manager](#) | [SSH client](#) | [EC2 serial console](#)

⚠ Port 22 (SSH) is open to all IPv4 addresses
 Port 22 (SSH) is currently open to all IPv4 addresses, indicated by **0.0.0.0/0** in the inbound rule in [your security group](#). For increased security, consider restricting access to only the EC2 Instance Connect service IP addresses for your Region: 18.206.107.24/29. [Learn more](#).

Instance ID
☒ i-0cad4d357fbf854d4 (student-attendance-server)

Connection Type

☒ **Connect using EC2 Instance Connect**
 Connect using the EC2 Instance Connect browser-based client, with a public IPv4 or IPv6 address.

☒ **Public IPv4 address**
☒ 44.214.108.132

☐ IPv6 address

☐ **Connect using EC2 Instance Connect Endpoint**
 Connect using the EC2 Instance Connect browser-based client, with a private IPv4 address and a VPC endpoint.

Username
 Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.

Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel Connect

14. In the Connect to instance page, click “connect” to directly access the console page.

```
#  
~\   ###      Amazon Linux 2  
~~ \_#####\  
~~  \###|     AL2 End of Life is 2025-06-30.  
~~   \#/        
~~    V~' '->    
~~~~          A newer version of Amazon Linux is available!  
~~.-.-         
~~/_/_/_/_/   Amazon Linux 2023, GA and supported until 2028-03-15.  
_/_/_/_/_/     https://aws.amazon.com/linux/amazon-linux-2023/  
[ec2-user@ip-10-0-7-137 ~]$  
  
[ec2-user@ip-10-0-7-137 ~]$ sudo yum update -y  
Loaded plugins: extras_suggestions, langpacks, priorities, update-motd  
amzn2-core | 3.6 kB 00:00:00  
No packages marked for update  
[ec2-user@ip-10-0-7-137 ~]$ sudo amazon-linux-extras install php8.0 mariadb10.5  
  
[ec2-user@ip-10-0-7-137 ~]$ sudo yum install -y httpd  
Loaded plugins: extras_suggestions, langpacks, priorities, update-motd  
Resolving Dependencies  
  
[ec2-user@ip-10-0-7-137 ~]$ sudo systemctl start httpd  
[ec2-user@ip-10-0-7-137 ~]$ sudo systemctl enable httpd  
Created symlink from /etc/systemd/system/multi-user.target.wants/httpd.service.  
[ec2-user@ip-10-0-7-137 ~]$ sudo usermod -a -G apache ec2-user  
  
[ec2-user@ip-10-0-7-137 ~]$ sudo usermod -a -G apache ec2-user  
[ec2-user@ip-10-0-7-137 ~]$ sudo chown -R ec2-user:apache /var/www  
[ec2-user@ip-10-0-7-137 ~]$ sudo chmod 2775 /var/www  
[ec2-user@ip-10-0-7-137 ~]$ find /var/www -type d -exec sudo chmod 2775 {} \  
[ec2-user@ip-10-0-7-137 ~]$ find /var/www -type f -exec sudo chmod 0664 {} \  
[ec2-user@ip-10-0-7-137 ~]$
```

```
sudo yum update -y
sudo amazon-linux-extras install php8.0 mariadb10.5
sudo yum install -y httpd
sudo systemctl start httpd
sudo systemctl enable httpd
sudo usermod -a -G apache ec2-user
//Refresh EC2 Instance Connect Page
sudo chown -R ec2-user:apache /var/www
sudo chmod 2775 /var/www
find /var/www -type d -exec sudo chmod 2775 {} \;
find /var/www -type f -exec sudo chmod 0664 {} \;
```

15. At the console page, type codes referred to the box above to connect to the Apache server and website.

i-0b68f9de95b55c0c9 (student-attendance-server)

Details | Status and alarms | Monitoring | Security | Networking

▼ Instance summary Info

Instance ID
i-0b68f9de95b55c0c9 (student-attendance-server)

Private IPv4 addresses
10.0.7.137

Instance state
Running

Public IPv4 address copied
44.202.116.68 | open address

IPv6 address
-

Public IPv4 DNS
ec2-44-202-116-68.compute-1.amazonaws.com

Test Page

This page is used to test the proper operation of the Apache HTTP server after it has been installed. If you can read this page, it means that the Apache HTTP server installed at this site is working properly.

If you are a member of the general public:

The fact that you are seeing this page indicates that the website you just visited is either experiencing problems, or is undergoing routine maintenance.

If you would like to let the administrators of this website know that you've seen this page instead of the page you expected, you should send them e-mail. In general, mail sent to the name "webmaster" and directed to the website's domain should reach the appropriate person.

For example, if you experienced problems while visiting `www.example.com`, you should send e-mail to "`webmaster@example.com`".

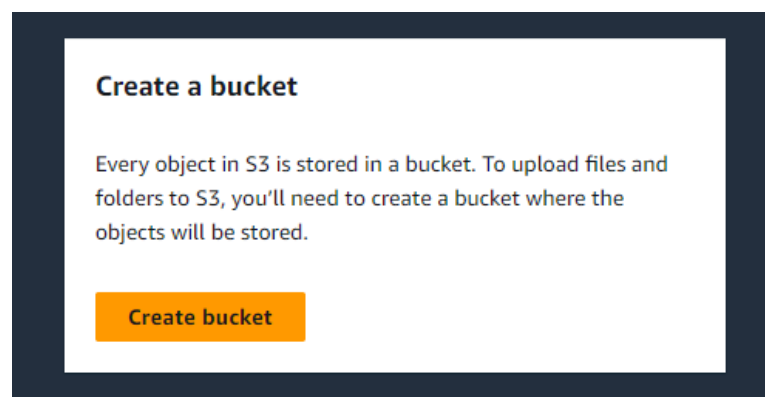
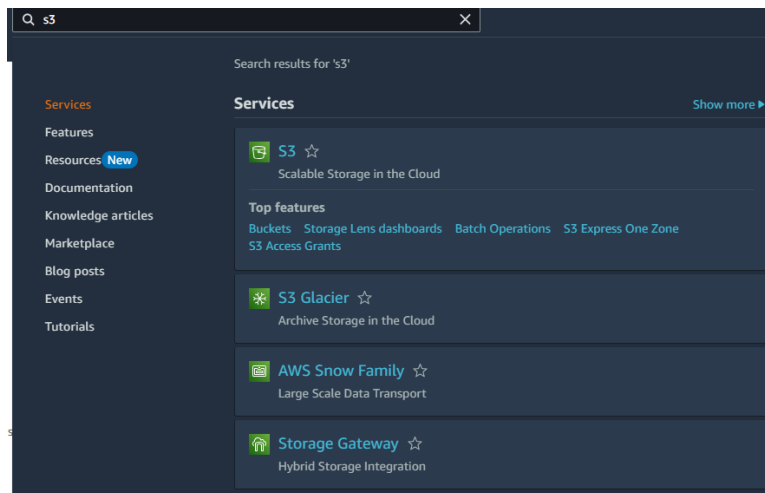
If you are the website administrator:

You may now add content to the directory `/var/www/html/`. Note that until you do so, people visiting your website will see this page, and not your content. To prevent this page from ever being used, follow the instructions in the file `/etc/httpd/conf.d/welcome.conf`.

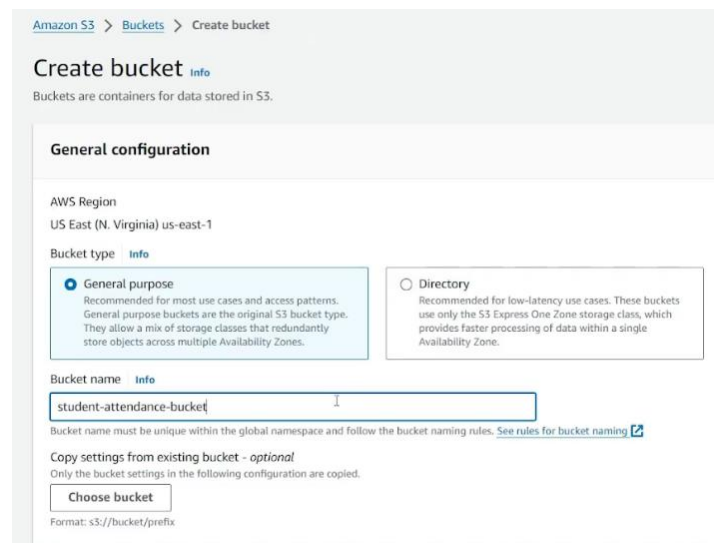
You are free to use the image below on web sites powered by the Apache HTTP Server:

Powered by **2.4**
APACHE

16. After the coding part, return to the instance page and copy the public IPv4 address. Then, paste it at the new tab to see if the website is working or not.



17. Then, go to S3 services to create a bucket to store the website file. Click “Create bucket”.



Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☐ **Block all public access**
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☐ **Block public access to buckets and objects granted through new access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
☐ **Block public access to buckets and objects granted through any access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.
☐ **Block public access to buckets and objects granted through new public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
☐ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

 **Turning off block all public access might result in this bucket and the objects within becoming public**
AWS recommends that you turn on block all public access, unless public access is required for specific and verified use cases such as static website hosting.
☒ I acknowledge that the current settings might result in this bucket and the objects within becoming public.

Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

☐ Disable
☒ Enable

▼ **Advanced settings**

Object Lock

Store objects using a write-once-read-many (WORM) model to help you prevent objects from being deleted or overwritten for a fixed amount of time or indefinitely. Object Lock works only in versioned buckets. [Learn more](#)

☒ Disable
☐ Enable

Permanently allows objects in this bucket to be locked. Additional Object Lock configuration is required in bucket details after bucket creation to protect objects in this bucket from being deleted or overwritten.

 Object Lock works only in versioned buckets. Enabling Object Lock automatically enables Versioning.

 After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.

Cancel **Create bucket**

18. In the Create bucket page, name the bucket name as a “student-attendance-bucket” and make sure the selected region is same as the selected region in the step. Then, make sure untick all the Block Public Access settings for this bucket and check the Default encryption. Click the “Create bucket”.

Amazon S3 > Buckets > student-attendance-bucket

student-attendance-bucket info

Objects | Properties | Permissions | Metrics | Management | Access Points

Permissions overview

Access finding
Access findings are provided by IAM external access analyzers. Learn more about [How IAM analyzer findings work](#).
[View analyzer for us-east-1](#)

Block public access (bucket settings)
Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

Bucket policy
The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

EditDelete

Bucket policy

EditDelete

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

No policy to display.

Copy

Bucket ARN

arn:aws:s3:::student-attendance-bucket

Policy

1 {
2 "Version": "2012-10-17",
3 "Statement": [
4 {
5 "Sid": "PublicReadGetObject",
6 "Effect": "Allow",
7 "Principal": "*",
8 "Action": [
9 "s3:GetObject"
10],
11 "Resource": [
12 "arn:aws:s3:::student-attendance-bucket/*"
13],
14 },
15],
16 }

Edit statement
PublicReadGetObject Remove

Add actions

Choose a service
Filter services

Included
S3

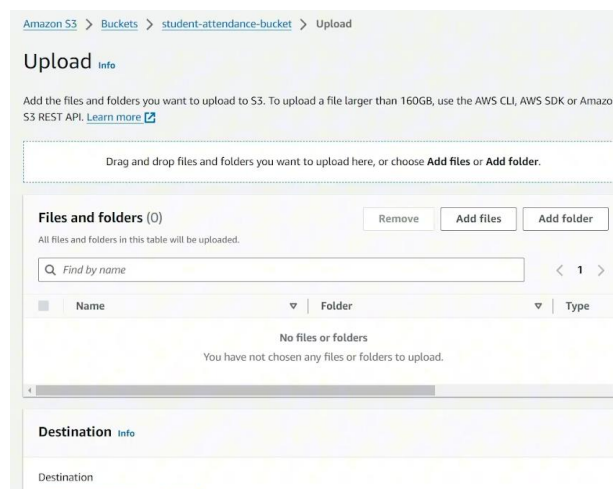
Available
AMP
API Gateway
API Gateway V2
ASC
Access Analyzer
Account

```

{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": [
        "s3:GetObject"
      ],
      "Resource": [
        "arn:aws:s3:::student-attendance-bucket/*"
        //arn: change to our own bucket.
      ]
    }
  ]
}

```

19. After bucket was created, click on the bucket name and go to permission tab. Edit the bucket policy and insert the code in the box above in the bucket policy.



Amazon S3 > Buckets > student-attendance-bucket > Upload

Upload info

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDK or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose [Add files](#) or [Add folder](#).

Files and folders (1 Total, 21.5 KB) Remove Add files Add folder

All files and folders in this table will be uploaded.

< 1 >

☐	Name	Folder	Type
☐	builtV21.zip	-	application/

Destination info

Destination

s3://student-attendance-bucket

Upload succeeded
View details below.

The information below will no longer be available after you navigate away from this page.

Summary

Destination s3://student-attendance-bucket	Succeeded 1 file, 21.5 KB (100.00%)	Failed 0 files, 0 B (0%)
---	--	-----------------------------

Files and folders | Configuration

Files and folders (1 Total, 21.5 KB)

< 1 >

Name	Folder	Type	Size	Status	Error
builtV21.zip	-	application/zip	21.5 KB	Succeeded	-

Properties | Permissions | Versions

Object overview

Owner awslabsc0w4870645t1669130846	S3 URI s3://student-attendance-bucket/builtV21.zip
AWS Region US East (N. Virginia) us-east-1	Amazon Resource Name (ARN) arn:aws:s3::student-attendance-bucket/builtV21.zip
Last modified October 1, 2024, 01:57:30 (UTC+08:00)	Entity tag (Etag) eb6ef77db07b16ca06afa5bafb6a953
Size 21.5 KB	Object URL Copied https://student-attendance-bucket.s3.amazonaws.com/builtV21.zip
Type zip	
Key builtV21.zip	

20. Next, go to Object tab and click “Add file” to upload the zip file. Put the file into the bucket. Click the detail of the file and copy the Object URL.

```
[ec2-user@ip-10-0-7-137 ~]$ mkdir stud-att
[ec2-user@ip-10-0-7-137 ~]$ ls
stud-att
[ec2-user@ip-10-0-7-137 ~]$ cd stud-att
```

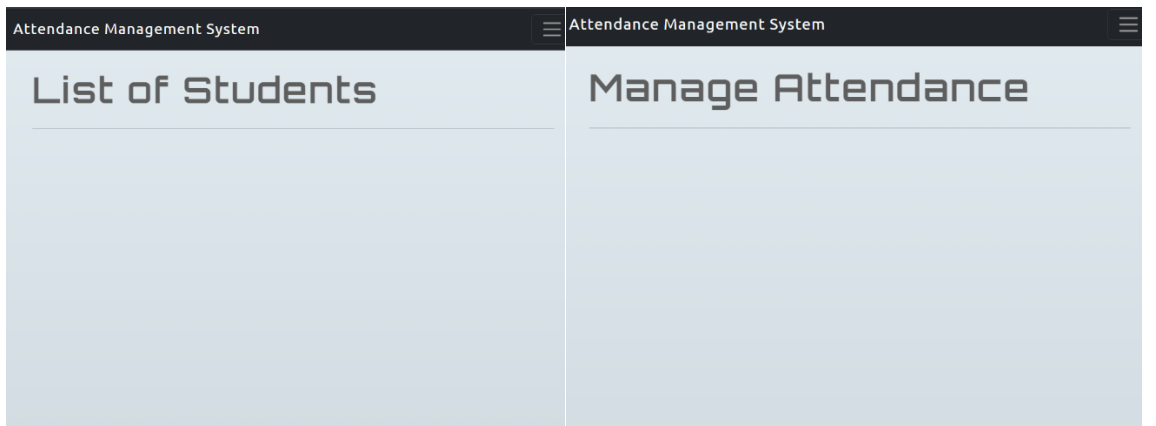
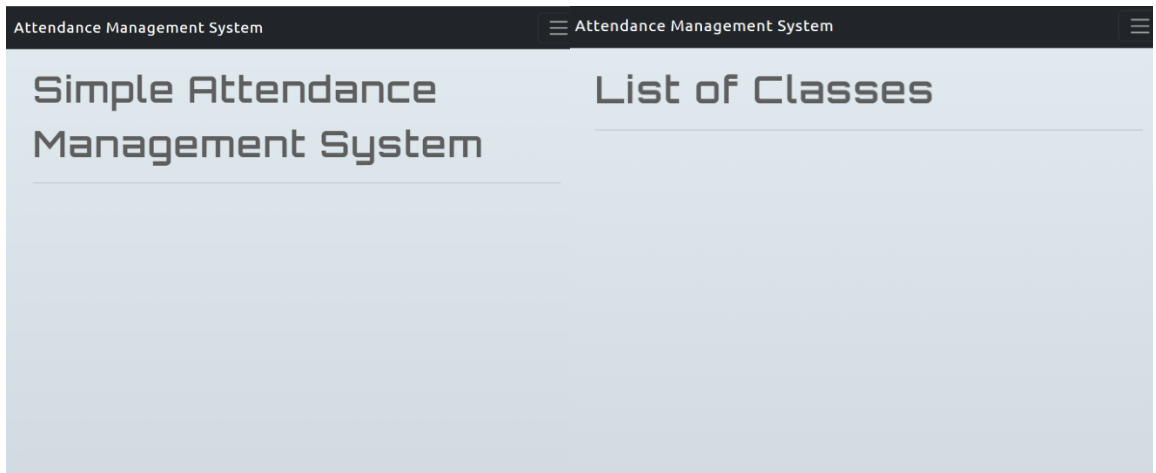
```
[ec2-user@ip-10-0-6-85 ~]$ cd stud-att
[ec2-user@ip-10-0-6-85 stud-att]$ wget https://student-attendance-bucket.s3.amazonaws.com/builtV21.zip
--2024-09-30 18:01:27-- https://student-attendance-bucket.s3.amazonaws.com/builtV21.zip
Resolving student-attendance-bucket.s3.amazonaws.com (student-attendance-bucket.s3.amazonaws.com)... 3.5.3.185, 52.217.192.201, 3.5.3.79, ...
Connecting to student-attendance-bucket.s3.amazonaws.com (student-attendance-bucket.s3.amazonaws.com)|3.5.3.185|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 22058 (22K) [application/zip]
Saving to: 'builtV21.zip'

100%[=====>] 22,058 --.-K/s in 0.001s

2024-09-30 18:01:27 (40.4 MB/s) - 'builtV21.zip' saved [22058/22058]
[ec2-user@ip-10-0-6-85 stud-att]$
```

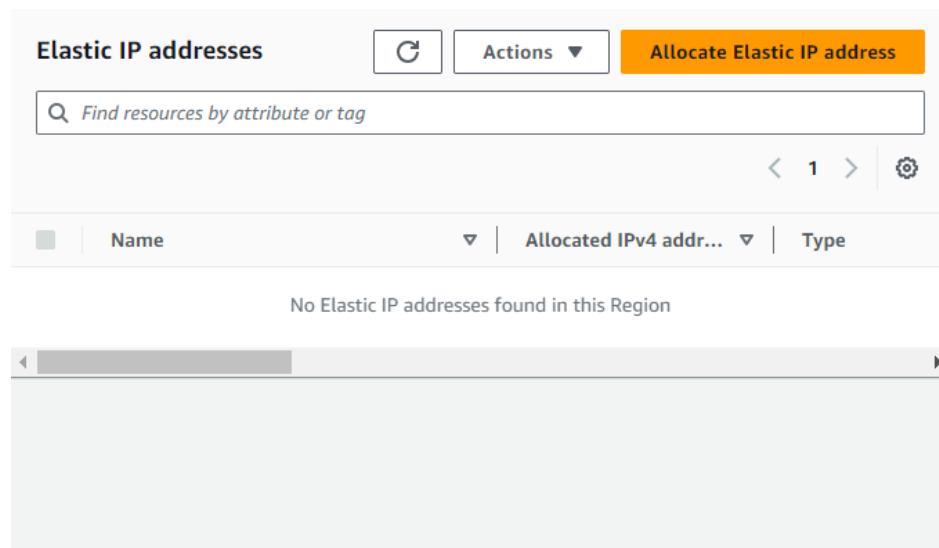
```
[ec2-user@ip-10-0-7-137 stud-att]$ ls
builtV21.zip
[ec2-user@ip-10-0-7-137 stud-att]$ unzip builtV21.zip
Archive:  builtV21.zip
  creating: built/
  inflating: built/ajax-api.php
  creating: built/assets/
  creating: built/assets/css/
  inflating: built/assets/css/style.css
  creating: built/assets/js/
  inflating: built/assets/js/script.js
  creating: built/classes/
  inflating: built/classes/actions.class.php
  creating: built/db/
  inflating: built/db-connect.php
  creating: built/inc/
  inflating: built/inc/footer.php
  inflating: built/inc/header.php
  inflating: built/inc/navigation.php
  inflating: built/index.php
  creating: built/modals/
  inflating: built/modals/class_form.php
  inflating: built/modals/settings-modal.php
  inflating: built/modals/student_form.php
  creating: built/pages/
  inflating: built/pages/attendance.php
  inflating: built/pages/attendance_report.php
  inflating: built/pages/class_list.php
  inflating: built/pages/home.php
  inflating: built/pages/student_list.php
  inflating: built/simple_attendance_db.sql
[ec2-user@ip-10-0-7-137 stud-att]$
```

```
Installing: built/simple_attendance_db.sql
[ec2-user@ip-10-0-7-137 stud-att]$ ls
builtV21.zip
[ec2-user@ip-10-0-7-137 stud-att]$ cd built
[ec2-user@ip-10-0-7-137 built]$ ls
ajax-api.php  classes  db-connect.php  index.php  pages
assets        db       inc             models     simple_attendance_db.sql
[ec2-user@ip-10-0-7-137 built]$ mv * /var/www/html
[ec2-user@ip-10-0-7-137 built]$ cd /var/www/html
[ec2-user@ip-10-0-7-137 html]$ ls
ajax-api.php  classes  db-connect.php  index.php  pages
assets        db       inc             models     simple_attendance_db.sql
```



```
mkdir stud-att
cd stud-att
wget https://student-attendance-bucket.s3.amazonaws.com/builtV21.zip
unzip builtV21.zip
cd built
mv * /var/www/html
cd /var/www/html
```

21. Return to the console page, create the directory using code “mkdir file” to store the code. Then, add the file inside the directory and unzip it. Next, move the file to ‘/var/www/html’ and check the website working or not. The output of the website does not have data inside on the test page on the list of classes, students, attendance and attendance report.



[EC2](#) > [Elastic IP addresses](#) > Allocate Elastic IP address

Allocate Elastic IP address [Info](#)

Elastic IP address settings [Info](#)

Public IPv4 address pool

- ☒ Amazon's pool of IPv4 addresses
 - ☐ Public IPv4 address that you bring to your AWS account with BYOIP. (option disabled because no pools found) [Learn more](#)
 - ☐ Customer-owned pool of IPv4 addresses created from your on-premises network for use with an Outpost. (option disabled because no customer owned pools found) [Learn more](#)
 - ☐ Allocate using an IPv4 IPAM pool (option disabled because no public IPv4 IPAM pools with AWS service as EC2 were found)

Network border group [Info](#)

Global static IP addresses

AWS Global Accelerator can provide global static IP addresses that are announced worldwide using anycast from AWS edge locations. This can help improve the availability and latency for your user traffic by using the Amazon global network. [Learn more](#)

Create accelerator

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tag

You can add up to 50 more tag

Cancel **Allocate**

22. Return to the instance page again and click “Allocate Elastic IP address”. Setup the Elastic IP address settings and click “Allocate”.

Elastic IP address allocated successfully.

Elastic IP address 3.92.118.127

Associate this Elastic IP address

Elastic IP addresses (1/1)

☒

Name

Allocated IPv4 addr...

Type

Allocation ID

Reverse DNS record

☒

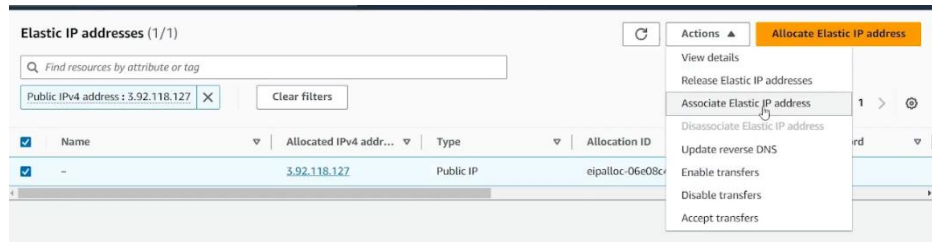
-

3.92.118.127

Public IP

eipalloc-06e08c42a617da6c4

-



Elastic IP address: 3.92.118.127

Resource type
Choose the type of resource with which to associate the Elastic IP address.

☒ Instance
☐ Network interface

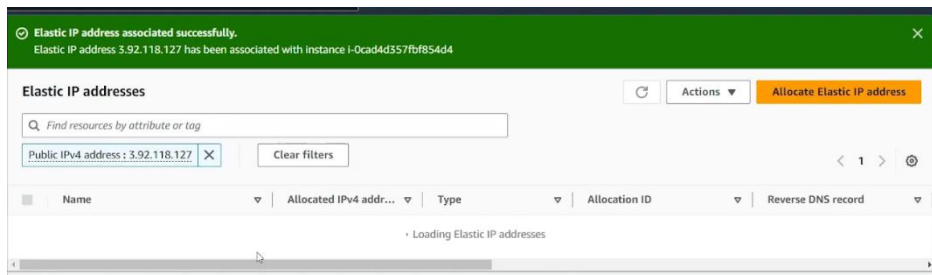
Instance
i-Ocad4d357fbf854d4

Private IP address
The private IP address with which to associate the Elastic IP address.
10.0.6.85

Reassociation
Specify whether the Elastic IP address can be reassociated with a different resource if it already associated with a resource.
☐ Allow this Elastic IP address to be reassociated

Cancel Associate

23. After allocated it, select the elastic IP address (10.0.6.85) and click ‘Associate’.





24. After we created the elastic IP, copy and paste the elastic ip address in the new tab to view the website. Now, we will create the database using the MariaDB monitor and Mysql engine.

```
[ec2-user@ip-10-0-7-137 ~]$ cd stud-att
[ec2-user@ip-10-0-7-137 stud-att]$ cd built
[ec2-user@ip-10-0-7-137 built]$ ls
[ec2-user@ip-10-0-7-137 built]$ cd /var/www/html
[ec2-user@ip-10-0-7-137 html]$ sudo mysql-u root
```

```
[ec2-user@ip-10-0-7-137 html]$ sudo systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; disabled; vendor preset: disabled)
   Active: inactive (dead)
     Docs: man:mariadbd(8)
           https://mariadb.com/kb/en/library/systemd/
[ec2-user@ip-10-0-7-137 html]$ sudo systemctl start mariadb
[ec2-user@ip-10-0-7-137 html]$ sudo systemctl enable mariadb
```

```
[ec2-user@ip-10-0-7-137 html]$ sudo mysql -u root
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.5.25-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> use mysql;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [mysql]> set password for 'root'@'localhost' = password('admin123');
Query OK, 0 rows affected (0.002 sec)

MariaDB [mysql]> quit;
Bye
```

```
[ec2-user@ip-10-0-7-137 html]$ sudo mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 5
Server version: 10.5.25-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE IF NOT EXISTS simple_attendance_db DEFAULT CHARACTER SET utf8mb4 COLLATE utf8mb4_general_ci;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> quit;
Bye
[ec2-user@ip-10-0-7-137 html]$ sudo mysql -h localhost -u root -p simple_attendance_db < simple_attendance_db.sql
Enter password:
[ec2-user@ip-10-0-7-137 html]$ ls
ajax-api.php  db-connect.php  index.php  simple_attendance_db.sql
[ec2-user@ip-10-0-7-137 html]$ nano config.php
```

```
GNU nano 2.9.8                                config.php

<?php
// Hostname
$host = "localhost";
// Username
$username = "root";
// Password
$pw = "admin123";
// Database Name
$dbname = "simple_attendance_db";

try{
    $conn = new MySQLi($host, $username, $pw, $dbname);
} catch(Exception $e){
    echo "Database Connection Failed: <br>";
    print_r($e->getMessage());
    exit;
}
?>
```

[Read 18 lines]

^G Get Help ^C Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos M-U Undo
^X Exit ^R Read File ^\ Replace ^U Uncut Text ^T To Spell ^_ Go To Line M-R Redo

i-0b68f9de95b55c0c9 (student-attendance-server)

PublicIPs: 98.83.176.36 PrivateIPs: 10.0.7.137

```
cd stud-att
cd built
cd /var/www/html
sudo systemctl status mariadb
sudo systemctl start mariadb
sudo systemctl enable mariadb
use mysql;
set password for 'root'@'localhost' = password('admin123');
quit;
```

```
sudo mysql -u root -p
```

```
CREATE DATABASE IF NOT EXISTS `simple_attendance_db` DEFAULT CHARACTER SET utf8mb4  
COLLATE utf8mb4_general_ci;
```

```
quit;
```

```
sudo mysql -h localhost -u root -p simple_attendance_db < simple_attendance_db.sql
```

```
nano config.php
```

```
<?php
```

```
// Hostname
```

```
$host = "localhost";
```

```
// Username
```

```
$uname = "root";
```

```
// Password
```

```
$pw = "admin123";
```

```
// Database Name
```

```
$dbname = "simple_attendance_db";
```

```
try{
```

```
    $conn = new mysqli ($servername, $username, $password, $dbname);
```

```
}catch(Exception $e){
```

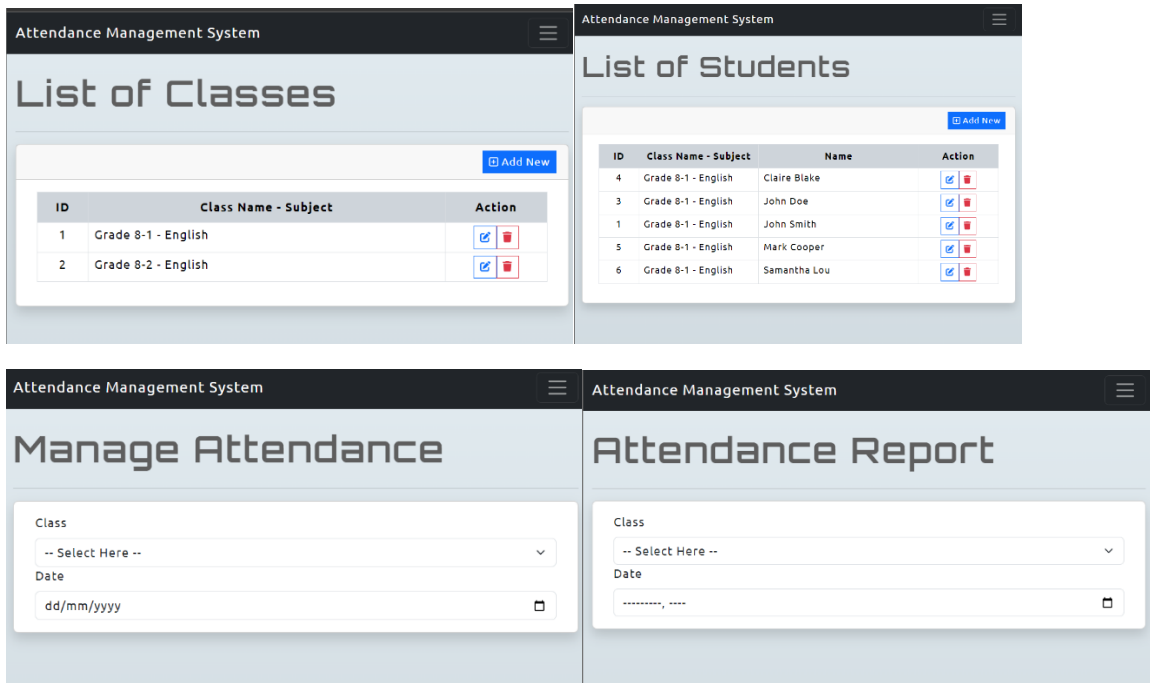
```
    echo "Database Connection Failed: <br>";
```

```
    print_r($e->getMessage());
```

```
    exit;
```

```
}
```

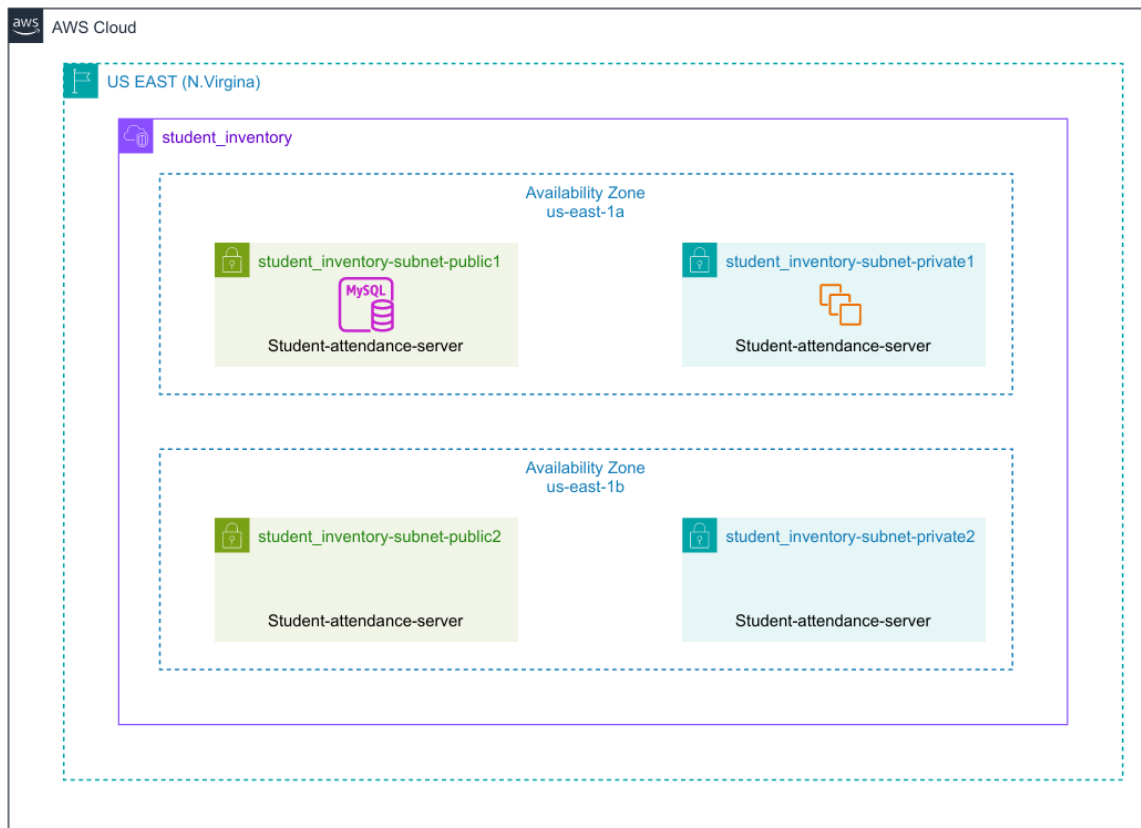
```
?>
```



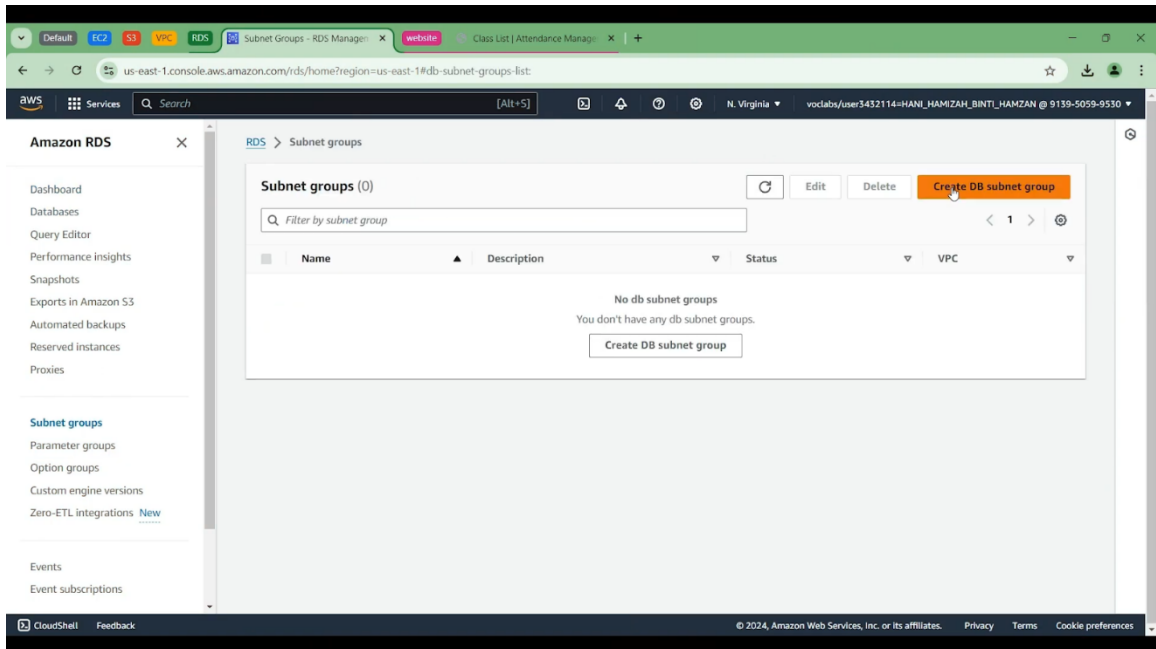
25. Return to console page again, create the database based on the code in the box above. This is the website output after database created.

2.0 Phase 2

2.1 Layout diagram of the revised network



2.2 Implementation steps



1. Search Amazon RDS and go to ‘Subnet groups’. Click “Create DB subnet group”.

RDS > Subnet groups > Create DB subnet group

Create DB subnet group

To create a new subnet group, give it a name and a description, and choose an existing VPC. You will then be able to add subnets related to that VPC.

Subnet group details

Name
You won't be able to modify the name after your subnet group has been created.

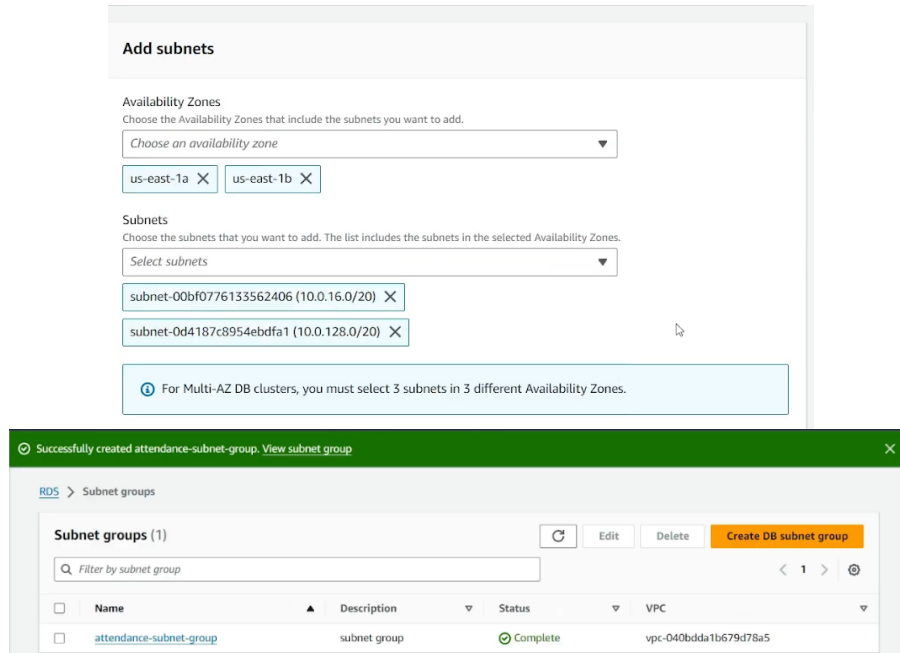
Must contain from 1 to 255 characters. Alphanumeric characters, spaces, hyphens, underscores, and periods are allowed.

Description

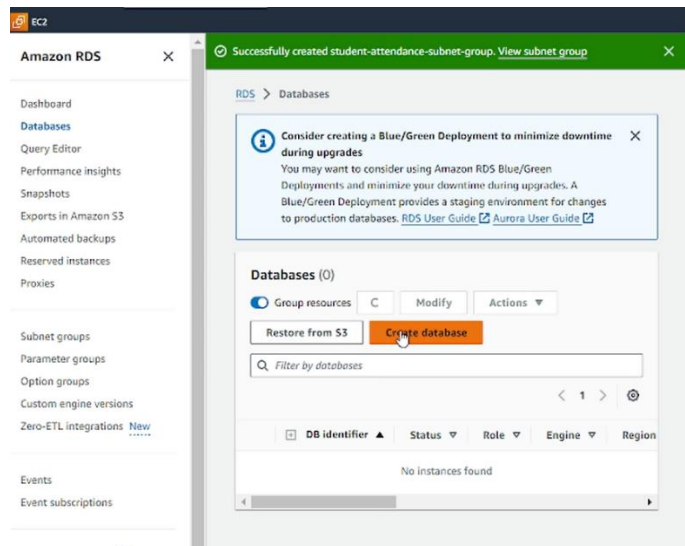
VPC
Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.

Add subnets

2. In a name, we put as a “attendance-subnet-group” and “subnet group” in the description . Fill up the description and select the VPC that we created in phase 1.



3. In the ‘Add subnet’ section, add two availability zones which are ‘us-east-1a’ and ‘us-east-1b’ and add its subnet too. Then, click “Create”. The new subnet that we created can see on the Subnet groups dashboard.



4. Next, go to Amazon RDS to create a RDS database and click “Create database”.

Successfully created student-attendance-subnet-group. [View subnet group](#)

RDS > Create database

Create database [Info](#)

Choose a database creation method

☒ **Standard create**
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ **Easy create**
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Engine options

Engine type [Info](#)

☐ Aurora (MySQL Compatible)


☐ Aurora (PostgreSQL Compatible)


☒ **MySQL**


☐ MariaDB


☐ PostgreSQL


☐ Oracle


☐ Microsoft SQL Server


☐ IBM Db2


Templates

Choose a sample template to meet your use case.

☐ **Production**
Use defaults for high availability and fast, consistent performance.

☐ **Dev/Test**
This instance is intended for development use outside of a production environment.

☒ **Free tier**
Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS. [Info](#)

5. Choose “standard create” for a database creation method and select “MySQL” for the engine type. Select “Free tier” for the templates.

Settings

DB instance identifier

Info

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

studentattendancedatabase

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 60 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

▼ Credentials Settings

Master username

Info

Type a login ID for the master user of your DB instance.

admin

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management

You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - most secure

RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed

Create your own password or have RDS create a password that you manage.

☐ Auto generate password

Amazon RDS can generate a password for you, or you can specify your own password.

Master password

Info

Password strength

Weak

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' " @

Confirm master password

Info

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class

Info

▼ Hide filters

☐ Show instance classes that support Amazon RDS Optimized Writes

Info

Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

☐ Include previous generation classes☐ Standard classes (includes m classes)☐ Memory optimized classes (includes r and x classes)☒ Burstable classes (includes t classes)

db.t4g.micro

6. In the ‘Settings’ section, we name the DB instance identifier as a “studentattendancedatabase” and select “Self managed”. Setup the master username as a “admin” and create our own master password which is “admin123”.

Connectivity
Info

Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☐ Don't connect to an EC2 compute resource
 Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☒ Connect to an EC2 compute resource
 Set up a connection to an EC2 compute resource for this database.

EC2 instance Info

Choose the EC2 instance to add as the compute resource for this database. A VPC security group is added to this EC2 instance. A VPC security group is also added to the database with an inbound rule that allows the EC2 instance to access the database.

i-0cad4d357fb854d4
student-attendance-server

Some VPC settings can't be changed when a compute resource is added

Adding an EC2 compute resource automatically selects the VPC, DB subnet group, and public access settings for this database. To allow the EC2 instance to access the database, a VPC security group `rds-ec2-X` is added to the database and another called `ec2-rds-X` to the EC2 instance. You can remove the new security group for the database only by removing the compute resource.

Virtual private cloud (VPC) Info

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

student-attendance-vpc (vpc-040bdda1b679d78a5)
2 Subnets, 2 Availability Zones

After a database is created, you can't change its VPC.

DB subnet group Info

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

☒ Choose existing
 Choose existing DB subnet group

☐ Automatic setup
 RDS creates a new subnet group for you or reuses an existing subnet group

Existing DB subnet groups

student-attendance-subnet-group
2 Subnets, 2 Availability Zones

Public access Info

☒ Yes

RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☐ No

RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) Info

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ Choose existing
 Choose existing VPC security groups

☐ Create new
 Create new VPC security group

Additional VPC security group

Choose one or more options

server-security-group
X

Amazon RDS will add a new VPC security group `rds-ec2-1` to allow connectivity with your compute resource.

Availability Zone Info

us-east-1a

Certificate authority - optional Info

Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

rds-ca-rsa2048-g1 (default)
Expiry: May 26, 2061

7. In the connectivity section, changed the Virtual private cloud (VPC) to the existing VPC, additional VPC security group select “server-security-group” and availability zone that we created.

▼ Additional configuration

Database options, encryption turned on, backup turned on, backtrack turned off, maintenance, CloudWatch Logs, delete protection turned off.

Database options

Initial database name [Info](#)

simple_attendance_db

If you do not specify a database name, Amazon RDS does not create a database.

DB parameter group [Info](#)

default.mysql8.0

Option group [Info](#)

default:mysql-8-0

Backup

☒ Enable automated backups

Creates a point-in-time snapshot of your database

Creating database studentattendancedatabase

View credential details

Your database might take a few minutes to launch. You can use settings from studentattendancedatabase to simplify configuration of suggested database add-ons while we finish creating your DB for you.

RDS > Databases

Consider creating a Blue/Green Deployment to minimize downtime during upgrades

You may want to consider using Amazon RDS Blue/Green Deployments and minimize your downtime during upgrades. A Blue/Green Deployment provides a staging environment for changes to production databases. [RDS User Guide](#) [Aurora User Guide](#)

Databases (1)

Group resources

Modify

Actions

Restore from S3

Create database

Filter by databases

DB identifier

Status

Role

Engine

Region ...

Size

Recommendations

studentattendancedatabase

Creating

Instance

MySQL Co...

us-east-1a

db.t4g.mi...

Successfully set up a connection between studentattendancedatabase and EC2 instance i-0cad4d357fbf854d4

Notifications 0 0 2 0 0 0

RDS > Databases

Consider creating a Blue/Green Deployment to minimize downtime during upgrades

You may want to consider using Amazon RDS Blue/Green Deployments and minimize your downtime during upgrades. A Blue/Green Deployment provides a staging environment for changes to production databases. [RDS User Guide](#) [Aurora User Guide](#)

Databases (1)

Group resources

Modify

Actions

Restore from S3

Create database

Filter by databases

DB identifier

Status

Role

Engine

Region ...

Size

Recommendations

studentattendancedatabase

Backing-up

Instance

MySQL Co...

us-east-1a

db.t4g.mi...

8. In the ‘Initial database name’, we named it as a “simple_attendance_db”. The creation of database took around 15 – 20 minutes to create.


```
aws Services Search [Alt+S] N. Virginia vodabs/user3432114=HANU_HAMIZAH_BINTI_HAMZAN @ 9139-5055

Keyboard shortcut
To tab out of the terminal window and select the next button element, press the left and right Shift keys together. Close permanently

ttendance_db.sql
[ec2-user@ip-10-0-6-85 html]$ sudo mysql -h studentattendancedatabase.ct6iudy5zur.us-east-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 30
Server version: 8.0.35 Source distribution

Copyright (c) 2000, 2019, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> ls
-> ;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'ls' at
line 1
MySQL [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| simple_attendance_db |
| sys |
+-----+
5 rows in set (0.004 sec)

MySQL [(none)]>
```

```
aws Services Search [Alt+S]

Keyboard shortcut
To tab out of the terminal window and select the next button element, press the left and right Shift keys together.

| simple_attendance_db |
| sys |
+-----+
5 rows in set (0.004 sec)

MySQL [(none)]> use simple_attendance_db
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MySQL [simple_attendance_db]> use simple_attendance_db;
Database changed
MySQL [simple_attendance_db]> show tables;
+-----+
| Tables in simple_attendance_db |
+-----+
| attendance_tbl |
| class_tbl |
| students_tbl |
+-----+
3 rows in set (0.002 sec)

MySQL [simple_attendance_db]> quit;
Bye
[ec2-user@ip-10-0-6-85 html]$ ls
ajax-api.php db-connect.php index.php simple_attendance_db.sql
[ec2-user@ip-10-0-6-85 html]$ nano config.php
[ec2-user@ip-10-0-6-85 html]$
```

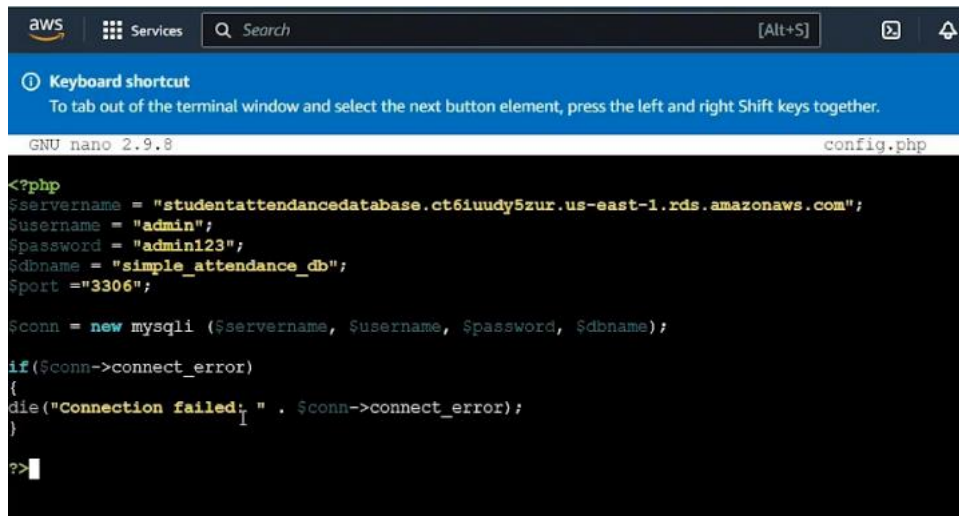
```
GNU nano 2.9.8 config.php

<?php
$servername = "student-attendancedb.cz6545r2fstt.us-east-1.rds.amazonaws.com";
$username = "admin";
$password = "admin123";
$dbname = "simple_attendance_db";
$port = "3306";

$conn = new mysqli ($servername, $username, $password, $dbname);

if($conn->connect_error)
{
die("Connection failed: " . $conn->connect_error);
}

?>
```



The screenshot shows an AWS console terminal window. At the top, there's a navigation bar with the AWS logo, 'Services', a search bar, and a '[Alt+S]' shortcut. Below this is a blue banner with a keyboard shortcut icon and text: 'Keyboard shortcut To tab out of the terminal window and select the next button element, press the left and right Shift keys together.' The terminal itself shows 'GNU nano 2.9.8' and 'config.php'. The code being edited is a PHP script for MySQL connection:

```
<?php
$servername = "studentattendancedatabase.ct6iuudy5zur.us-east-1.rds.amazonaws.com";
$username = "admin";
$password = "admin123";
$dbname = "simple_attendance_db";
$port = "3306";

$conn = new mysqli ($servername, $username, $password, $dbname);

if($conn->connect_error)
{
    die("Connection failed: " . $conn->connect_error);
}

?>
```

```
cd stud-att
```

```
ls
```

```
cd built
```

```
ls
```

```
cd /var/www/html
```

```
ls
```

```
sudo mysql -h studentattendancedatabase.cwoqz7r06eud.us-east-1.rds.amazonaws.com -u
admin -padmin123 simple_attendance_db < simple_attendance_db.sql
```

```
admin123
```

```
show databases;

use simple_attendance_db;

show tables;

quit;

ls

nano config.php

<?php

$servername = "studentattendancedatabase.cwoqz7r06eud.us-east-1.rds.amazonaws.com";

$username = "admin";

$password = "admin123";

$dbname = "simple_attendance_db";

$port = "3306";


$conn = new mysqli ($servername, $username, $password, $dbname);


if($conn->connect_error)
{
die("Connection failed: " . $conn->connect_error);
}

?>
```

10. In the console page, type the code referred to the box above to create a database.

```
[ec2-user@ip-10-0-7-252 html]$ ls
ajax-api.php  config.php  db-connect.php  index.php  simple_attendance_db.
[ec2-user@ip-10-0-7-252 html]$ nano db-connect.php
```

```

<?php
// Hostname
$host = "studentattendancedatabase.ct6iuudy5zur.us-east-1.rds.amazonaws.com";
// Username
$username = "admin";
// Password
$pw = "admin123";
// Database Name
$dbname = "simple_attendance_db";

try{
    $conn = new MySQLi($host, $username, $pw, $dbname);
} catch(Exception $e){
    echo "Database Connection Failed: <br>";
    print_r($e->getMessage());
    exit;
}
?>

```

```

MySQL [simple_attendance_db]> show tables;
+-----+
| Tables_in_simple_attendance_db |
+-----+
| attendance_tbl                  |
| class_tbl                      |
| students_tbl                   |
+-----+
3 rows in set (0.001 sec)

MySQL [simple_attendance_db]> select * from class_tbl;
+-----+-----+-----+-----+
| id | name                | created_at          | updated_at          |
+-----+-----+-----+-----+
| 1  | Grade 8-1 - English | 2023-11-16 11:37:26 | 2023-11-16 11:52:34 |
| 2  | Grade 8-2 - English | 2023-11-16 11:52:46 | NULL                |
+-----+-----+-----+-----+
2 rows in set (0.001 sec)

```

Attendance Management System

New Class has been added successfully!

List of Classes

Add New

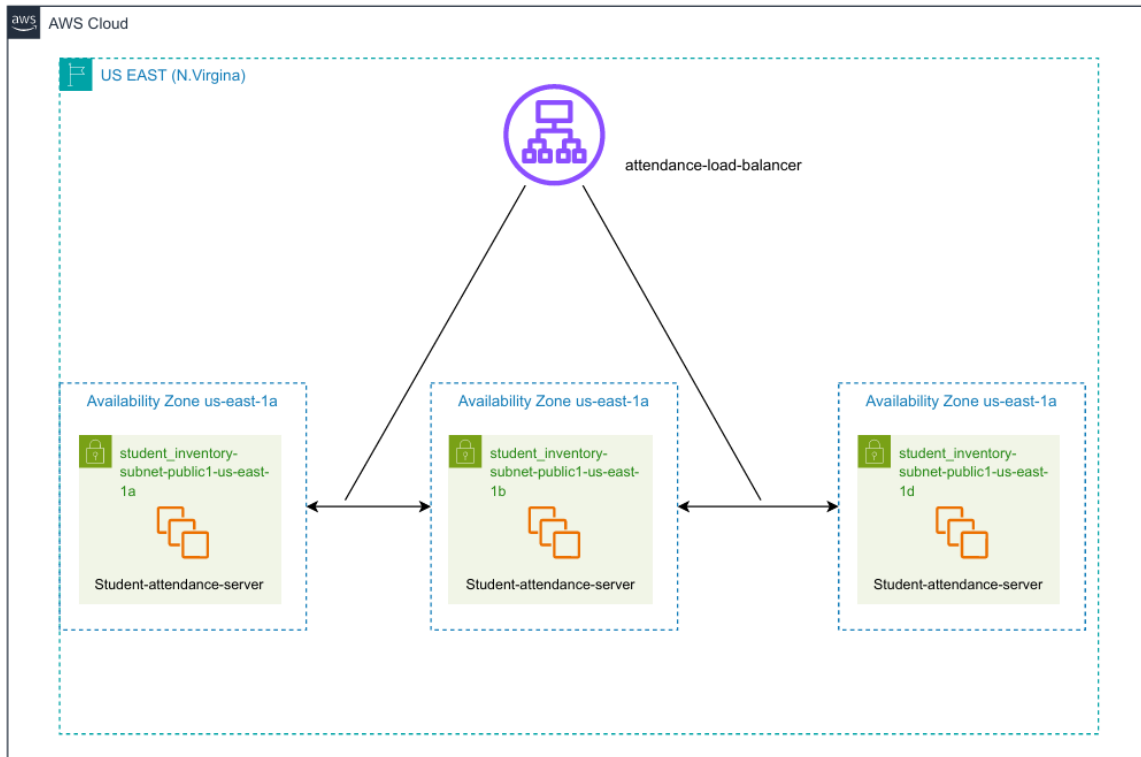
ID	Class Name - Subject	Action
1	Grade 8-1 - English	 
2	Grade 8-2 - English	 
4	Grade 8-5 - Math	 

```
MySQL [simple_attendance_db]> select * from class_tbl;
+----+-----+-----+-----+
| id | name                | created_at          | updated_at          |
+----+-----+-----+-----+
| 1  | Grade 8-1 - English | 2023-11-16 11:37:26 | 2023-11-16 11:52:34 |
| 2  | Grade 8-2 - English | 2023-11-16 11:52:46 | NULL                |
| 4  | Grade 8-5 - Math    | 2024-10-02 22:25:37 | NULL                |
+----+-----+-----+-----+
3 rows in set (0.001 sec)
```

11. Type “nano db-connect.php” to connect the database. Then, paste the code to the db-connect.php. This is when you make changes like add/edit/update anything from the website, the database also updated. We can view it by typing command select * from the tables that we wanted. For example, we add new class ‘Grade 8-5 - Math’, it shows new class has been added successfully in our website. Go back to console, it showed that the data is connected and updated from our website.

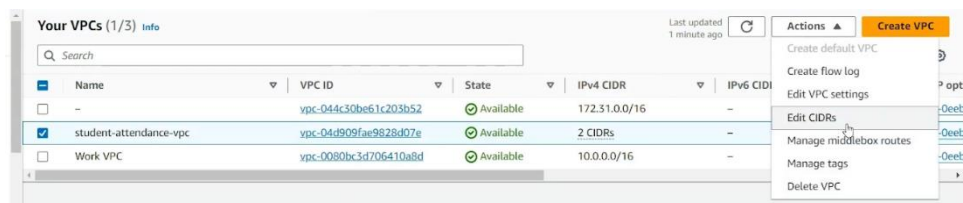
3.0 Phase 3

3.1 Layout diagram of the revised network



3.2 Implementation steps

In this phase, we will do the decoupling. We need to create a public subnet first then we will create an elastic load balance.



1. In the VPC dashboard, select the VPC that we created and go to 'Actions'. Choose 'Edit CIDRs'.

Add IPv4 CIDR

IPv4 CIDR block

☒ IPv4 CIDR manual input
 ☐ IPAM-allocated IPv4 CIDR

IPv4 CIDR

20.0.0.0/16

CIDR block size must be between /16 and /28.

Cancel

Save

2. Click ‘Add new IPv4 CIDR’ and fill up the IPv4 CIDR with . Click “Save”.

Edit CIDRs

Add or remove CIDR blocks for your VPC.

IPv4 CIDRs

CIDR	Status	
10.0.0.0/16	Associated	Remove
20.0.0.0/16	Associated	Remove

Add new IPv4 CIDR

IPv6 CIDRs

CIDR (Network border group)	Pool	Address attribute	Status
You have no IPv6 CIDR blocks associated with your VPC.			

Add new IPv6 CIDR

Close

3. After that, we can view the changes that we made in the IPv4 CIDRs dashboard. Click “Close” and return to “Your VPCs” page.

Your VPCs (1/3)

Last updated less than a minute ago

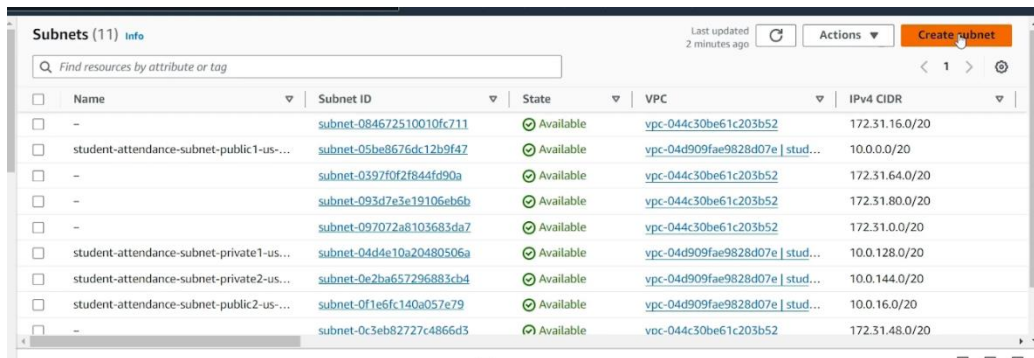
Actions

Create VPC

Search

	Name	VPC ID	State	IPv4 CIDR	IPv6 CIDR	DHCP opti
☐	-	vpc-044c30be61c203b52	Available	172.31.0.0/16	-	dopt-0eeb
☒	student-attendance-vpc	vpc-04d909fae9828d07e	Available	2 CIDRs	-	dopt-0eeb
☐	Work VPC	vpc-0080bc3d706410a8d	Available	10.0.0.0/16	-	dopt-0eeb

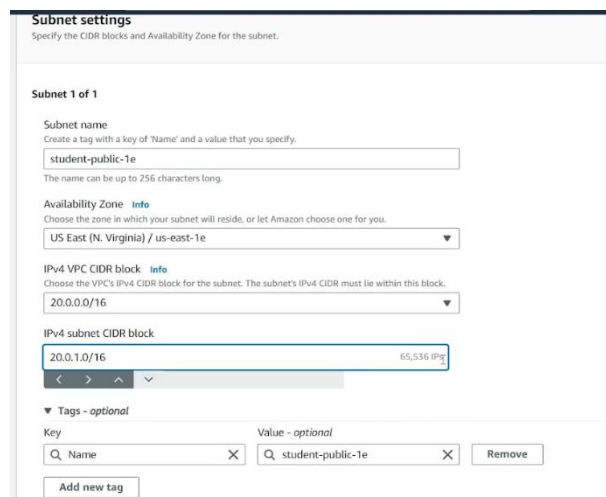
4. In the VPC table, it shows IPv4 CIDR has 2 CIDRs.



The screenshot shows the AWS Subnets console with a table of 11 subnets. The table has columns for Name, Subnet ID, State, VPC, and IPv4 CIDR. The subnets are listed in descending order of creation time. The first subnet is 'student-attendance-subnet-public-1-us-east-1e' with Subnet ID 'subnet-084672510010fc711' and IPv4 CIDR '172.31.16.0/20'. The second subnet is 'student-attendance-subnet-private-1-us-east-1e' with Subnet ID 'subnet-05be8676dc12b9f47' and IPv4 CIDR '10.0.0.0/20'. The third subnet is 'student-attendance-subnet-private-2-us-east-1e' with Subnet ID 'subnet-0397f0f2f844fd90a' and IPv4 CIDR '172.31.64.0/20'. The fourth subnet is 'student-attendance-subnet-public-2-us-east-1e' with Subnet ID 'subnet-093d7e3e19106eb6b' and IPv4 CIDR '172.31.80.0/20'. The fifth subnet is 'student-attendance-subnet-private-3-us-east-1e' with Subnet ID 'subnet-097072a8103683da7' and IPv4 CIDR '172.31.0.0/20'. The sixth subnet is 'student-attendance-subnet-private-4-us-east-1e' with Subnet ID 'subnet-04d4e10a20480506a' and IPv4 CIDR '10.0.128.0/20'. The seventh subnet is 'student-attendance-subnet-private-5-us-east-1e' with Subnet ID 'subnet-0e2ba657296883cb4' and IPv4 CIDR '10.0.144.0/20'. The eighth subnet is 'student-attendance-subnet-private-6-us-east-1e' with Subnet ID 'subnet-0f1e6fc140a057e79' and IPv4 CIDR '10.0.16.0/20'. The ninth subnet is 'student-attendance-subnet-private-7-us-east-1e' with Subnet ID 'subnet-0c3eb82727c4866d3' and IPv4 CIDR '172.31.48.0/20'. The tenth subnet is 'student-attendance-subnet-private-8-us-east-1e' with Subnet ID 'subnet-0c3eb82727c4866d3' and IPv4 CIDR '172.31.48.0/20'. The eleventh subnet is 'student-attendance-subnet-private-9-us-east-1e' with Subnet ID 'subnet-0c3eb82727c4866d3' and IPv4 CIDR '172.31.48.0/20'.

Name	Subnet ID	State	VPC	IPv4 CIDR
-	subnet-084672510010fc711	Available	vpc-044c30be61c203b52	172.31.16.0/20
student-attendance-subnet-public-1-us-east-1e	subnet-05be8676dc12b9f47	Available	vpc-04d909fae9828d07e stud...	10.0.0.0/20
-	subnet-0397f0f2f844fd90a	Available	vpc-044c30be61c203b52	172.31.64.0/20
-	subnet-093d7e3e19106eb6b	Available	vpc-044c30be61c203b52	172.31.80.0/20
-	subnet-097072a8103683da7	Available	vpc-044c30be61c203b52	172.31.0.0/20
student-attendance-subnet-private-1-us-east-1e	subnet-04d4e10a20480506a	Available	vpc-04d909fae9828d07e stud...	10.0.128.0/20
student-attendance-subnet-private-2-us-east-1e	subnet-0e2ba657296883cb4	Available	vpc-04d909fae9828d07e stud...	10.0.144.0/20
student-attendance-subnet-public-2-us-east-1e	subnet-0f1e6fc140a057e79	Available	vpc-04d909fae9828d07e stud...	10.0.16.0/20
-	subnet-0c3eb82727c4866d3	Available	vpc-044c30be61c203b52	172.31.48.0/20
-	subnet-0c3eb82727c4866d3	Available	vpc-044c30be61c203b52	172.31.48.0/20
-	subnet-0c3eb82727c4866d3	Available	vpc-044c30be61c203b52	172.31.48.0/20

5. In the left bar, go to subnets and click “Create subnet” to create a public subnet. For the VPC, choose the VPC that we created in phase 1.



The screenshot shows the AWS Subnet settings form. The form is titled 'Subnet settings' and has a subtitle 'Specify the CIDR blocks and Availability Zone for the subnet.' The form is divided into several sections. The first section is 'Subnet 1 of 1'. The second section is 'Subnet name'. The third section is 'Availability Zone'. The fourth section is 'IPv4 VPC CIDR block'. The fifth section is 'IPv4 subnet CIDR block'. The sixth section is 'Tags - optional'.

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
student-public-1e
The name can be up to 256 characters long.

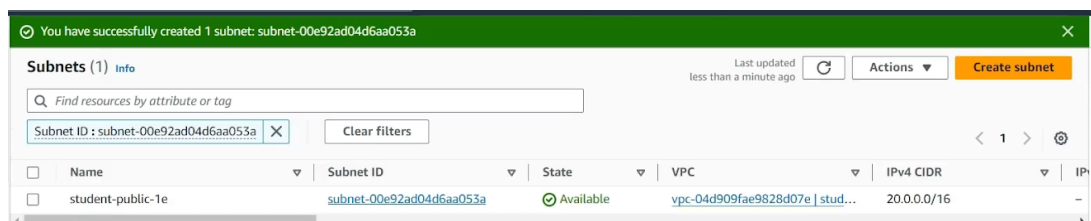
Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
US East (N. Virginia) / us-east-1e

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
20.0.0.0/16

IPv4 subnet CIDR block
20.0.1.0/16 65,536 IP
< > ^ v

Tags - optional
Key Value - optional
Name student-public-1e Remove
Add new tag

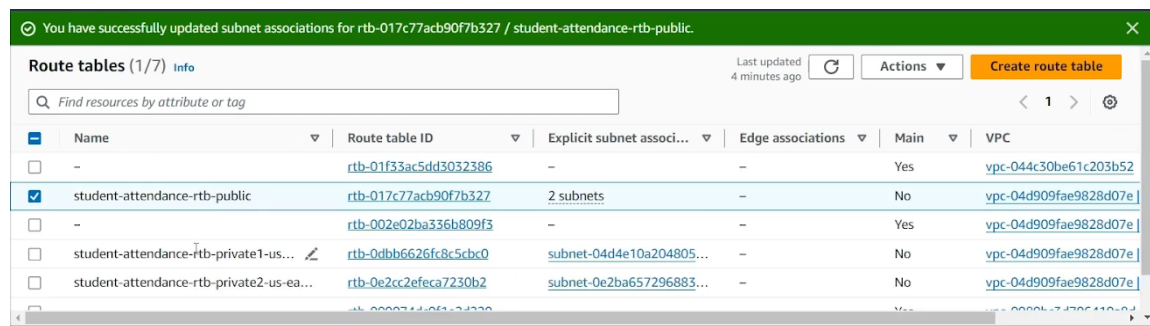
6. In Subnet settings, filled up the Subnet name as a “student-public-1e”. Choose the availability zone and put the IPv4 CIDR block that we created just now. Then, click “Create subnet”.



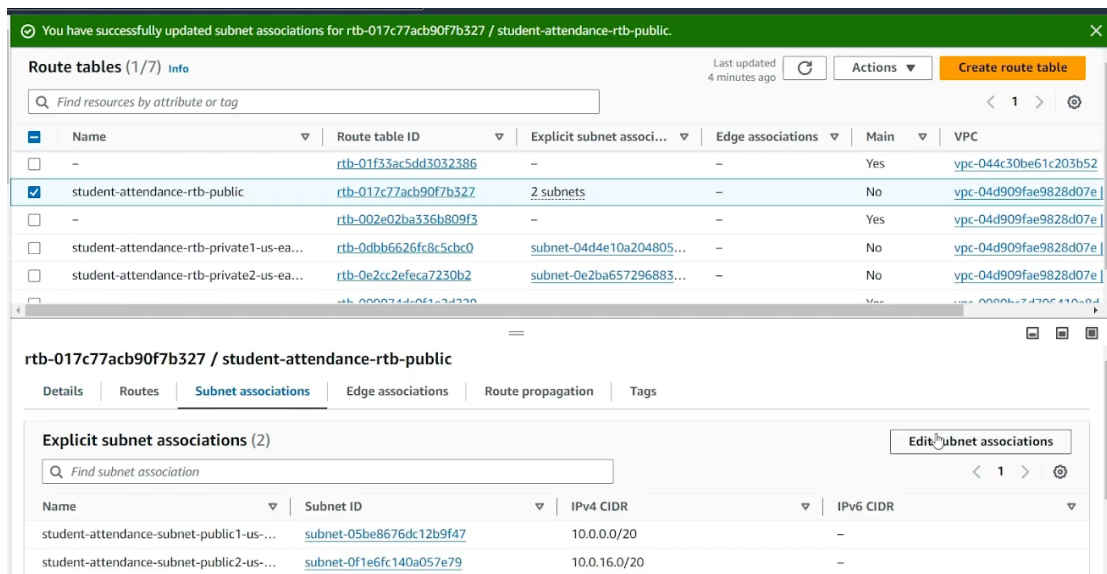
The screenshot shows the AWS Subnets console with a notification at the top: 'You have successfully created 1 subnet: subnet-00e92ad04d6aa053a'. Below the notification is a table of subnets. The table has columns for Name, Subnet ID, State, VPC, and IPv4 CIDR. The first subnet is 'student-public-1e' with Subnet ID 'subnet-00e92ad04d6aa053a' and IPv4 CIDR '20.0.0.0/16'. The second subnet is 'student-attendance-subnet-public-1-us-east-1e' with Subnet ID 'subnet-084672510010fc711' and IPv4 CIDR '172.31.16.0/20'. The third subnet is 'student-attendance-subnet-private-1-us-east-1e' with Subnet ID 'subnet-05be8676dc12b9f47' and IPv4 CIDR '10.0.0.0/20'. The fourth subnet is 'student-attendance-subnet-private-2-us-east-1e' with Subnet ID 'subnet-0397f0f2f844fd90a' and IPv4 CIDR '172.31.64.0/20'. The fifth subnet is 'student-attendance-subnet-private-3-us-east-1e' with Subnet ID 'subnet-093d7e3e19106eb6b' and IPv4 CIDR '172.31.80.0/20'. The sixth subnet is 'student-attendance-subnet-private-4-us-east-1e' with Subnet ID 'subnet-097072a8103683da7' and IPv4 CIDR '172.31.0.0/20'. The seventh subnet is 'student-attendance-subnet-private-5-us-east-1e' with Subnet ID 'subnet-04d4e10a20480506a' and IPv4 CIDR '10.0.128.0/20'. The eighth subnet is 'student-attendance-subnet-private-6-us-east-1e' with Subnet ID 'subnet-0e2ba657296883cb4' and IPv4 CIDR '10.0.144.0/20'. The ninth subnet is 'student-attendance-subnet-private-7-us-east-1e' with Subnet ID 'subnet-0f1e6fc140a057e79' and IPv4 CIDR '10.0.16.0/20'. The tenth subnet is 'student-attendance-subnet-private-8-us-east-1e' with Subnet ID 'subnet-0c3eb82727c4866d3' and IPv4 CIDR '172.31.48.0/20'. The eleventh subnet is 'student-attendance-subnet-private-9-us-east-1e' with Subnet ID 'subnet-0c3eb82727c4866d3' and IPv4 CIDR '172.31.48.0/20'.

Name	Subnet ID	State	VPC	IPv4 CIDR
student-public-1e	subnet-00e92ad04d6aa053a	Available	vpc-04d909fae9828d07e stud...	20.0.0.0/16
-	subnet-084672510010fc711	Available	vpc-044c30be61c203b52	172.31.16.0/20
student-attendance-subnet-public-1-us-east-1e	subnet-05be8676dc12b9f47	Available	vpc-04d909fae9828d07e stud...	10.0.0.0/20
-	subnet-0397f0f2f844fd90a	Available	vpc-044c30be61c203b52	172.31.64.0/20
-	subnet-093d7e3e19106eb6b	Available	vpc-044c30be61c203b52	172.31.80.0/20
-	subnet-097072a8103683da7	Available	vpc-044c30be61c203b52	172.31.0.0/20
student-attendance-subnet-private-1-us-east-1e	subnet-04d4e10a20480506a	Available	vpc-04d909fae9828d07e stud...	10.0.128.0/20
student-attendance-subnet-private-2-us-east-1e	subnet-0e2ba657296883cb4	Available	vpc-04d909fae9828d07e stud...	10.0.144.0/20
student-attendance-subnet-public-2-us-east-1e	subnet-0f1e6fc140a057e79	Available	vpc-04d909fae9828d07e stud...	10.0.16.0/20
-	subnet-0c3eb82727c4866d3	Available	vpc-044c30be61c203b52	172.31.48.0/20
-	subnet-0c3eb82727c4866d3	Available	vpc-044c30be61c203b52	172.31.48.0/20
-	subnet-0c3eb82727c4866d3	Available	vpc-044c30be61c203b52	172.31.48.0/20

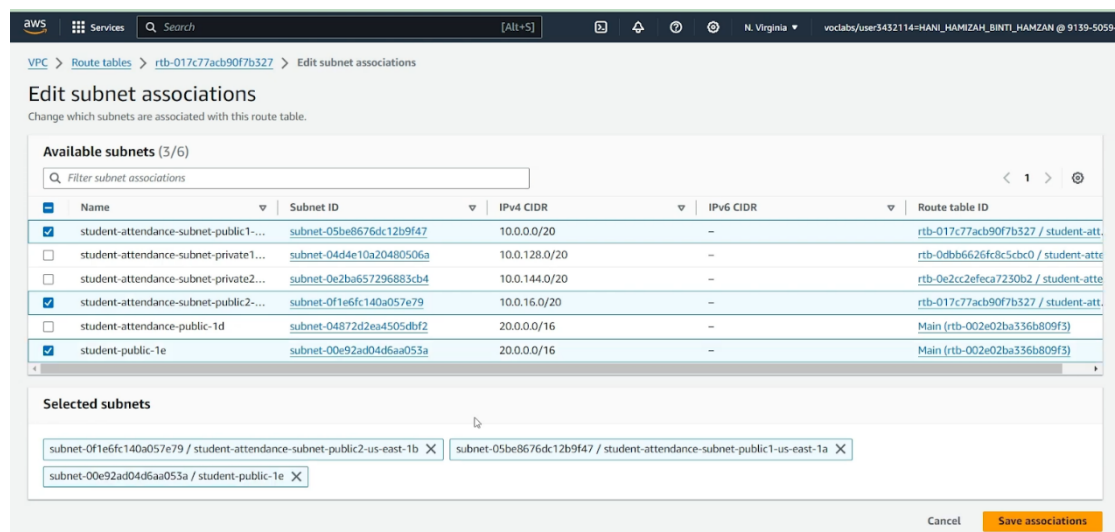
7. After creating the subnet, make sure the subnet state status is ‘Available’.



8. In the left bar, go to “Route tables” page.



9. Select the route tables with 2 subnets. Scroll down to the details of the route tables and go to “Subnet associations” tab. Then, click “Edit subnet associations”.



10. Select all the public subnets and click “Save associations”.

You have successfully updated subnet associations for rtb-017c77acb90f7b327 / student-attendance-rtb-public.

Route tables (1/7) [Info](#)

Find resources by attribute or tag

Last updated less than a minute ago [Actions](#) [Create route table](#)

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	-	rtb-01f33ac5dd3032386	-	-	Yes	vpc-044c30be61c203b52
☒	student-attendance-rtb-public	rtb-017c77acb90f7b327	3 subnets	-	No	vpc-04d909fae9828d07e
☐	-	rtb-002c02ba3336b809f3	-	-	Yes	vpc-04d909fae9828d07e
☐	student-attendance-rtb-private1-us-ea...	rtb-0dbb6626f8c5cb0	subnet-04d4e10a204805...	-	No	vpc-04d909fae9828d07e
☐	student-attendance-rtb-private2-us-ea...	rtb-0e2cc2efeca7230b2	subnet-0e2ba657296883...	-	No	vpc-04d909fae9828d07e

rtb-017c77acb90f7b327 / student-attendance-rtb-public

[Details](#) [Routes](#) [Subnet associations](#) [Edge associations](#) [Route propagation](#) [Tags](#)

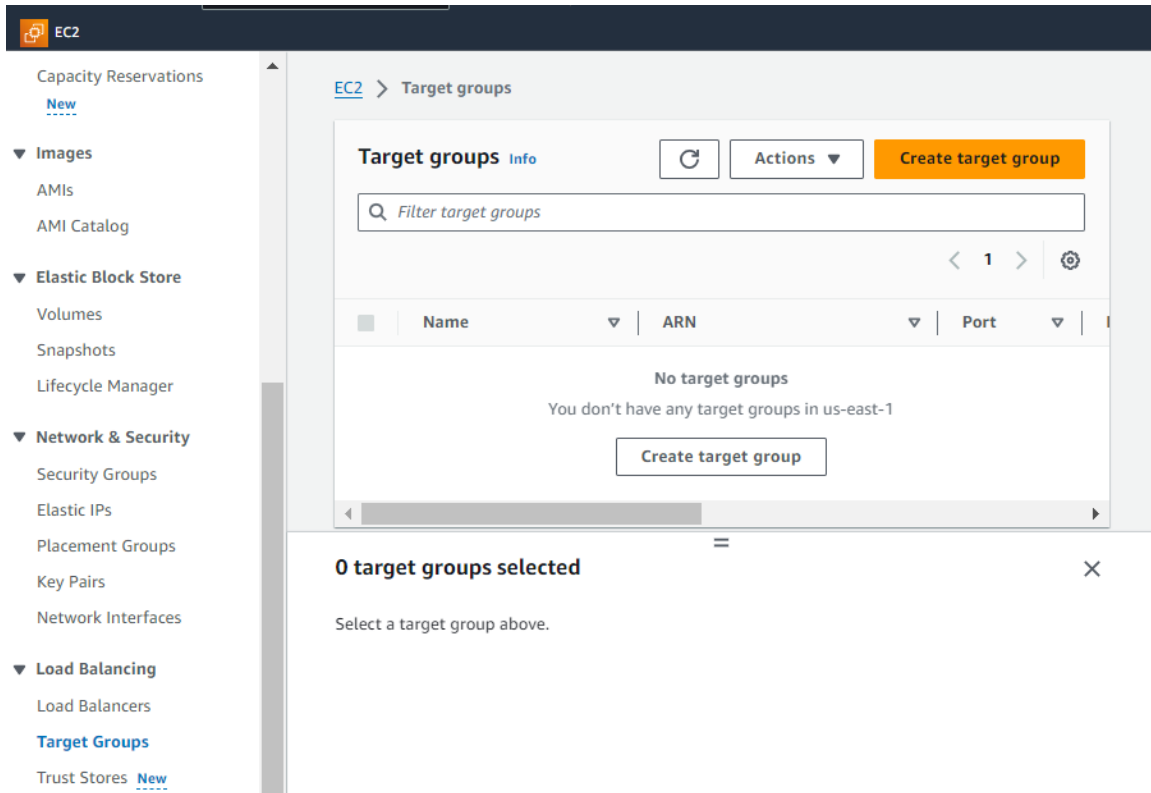
Details

Route table ID rtb-017c77acb90f7b327	Main No	Explicit subnet associations 3 subnets	Edge associations -
VPC vpc-04d909fae9828d07e student-attendance-vpc	Owner ID 913950599530		

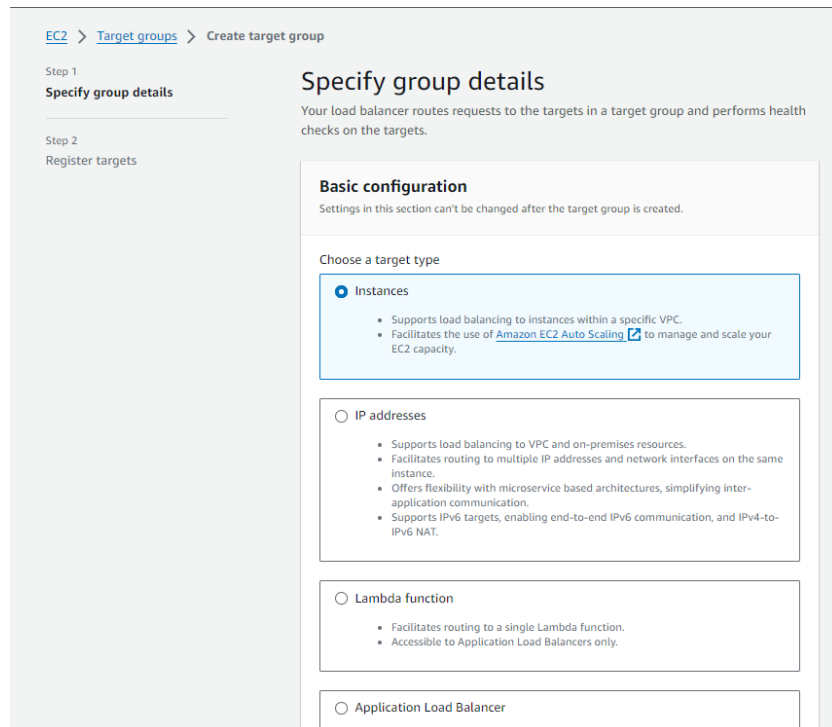
11. After save the progress, return to Route tables and the selected route has 3 subnets.

ELB created on a new AZ

Now, we will create a load balancing target group. Under the ‘Load Balancing’ in EC2, go to ‘Target Groups’.



12. In the Target groups page, click “Create target group”.



13. Choose ‘Instances’ as a target type.

- Facilitates routing to multiple IP addresses and network interfaces on the same instance.
- Offers flexibility with microservice based architectures, simplifying inter-application communication.
- Supports IPv6 targets, enabling end-to-end IPv6 communication, and IPv4-to-IPv6 NAT.

☐ Lambda function

- Facilitates routing to a single Lambda function.
- Accessible to Application Load Balancers only.

☐ Application Load Balancer

- Offers the flexibility for a Network Load Balancer to accept and route TCP requests within a specific VPC.
- Facilitates using static IP addresses and PrivateLink with an Application Load Balancer.

Target group name

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Protocol : Port

Choose a protocol for your target group that corresponds to the Load Balancer type that will route traffic to it. Some protocols now include anomaly detection for the targets and you can set mitigation options once your target group is created. This choice cannot be changed after creation

HTTP

80

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4

Each instance has a default network interface (eth0) that is assigned the primary private IPv4

☐ IPv6

Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

student-attendance-vpc

vpc-04d909fae9828d07e

IPv4 VPC CIDR: 10.0.0.0/16

Protocol version

☒ HTTP1

Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

☐ HTTP2

Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

☐ gRPC

Send requests to targets using gRPC. Supported when the request protocol is gRPC.

14. Fill up the Target group name as a “student-attendance-target-group” and select “HTTP” for Protocol : Port. Select “IPv4” for the IP address type and choose “student-attendance-vpc” for the VPC that we created in phase 1.

Health checks

The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol

HTTP

Health check path

Use the default path of “/” to perform health checks on the root, or specify a custom path if preferred.

Up to 1024 characters allowed.

15. In Health checks, select “HTTP” in ‘Health check protocol’ and fill up “/index.php” for the ‘Health check path’. Then, click ‘Next’.

Register targets

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (1/1)

Filter instances

☒	Instance ID	Name	State	Security groups
☒	i-09dc6574d7842ee2b	student-attendance-server	Running	ec2-rds-1, server-security-group

1 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

Include as pending below

16. In the Available instances, select the available instances and click “Include as pending below”.

Review targets

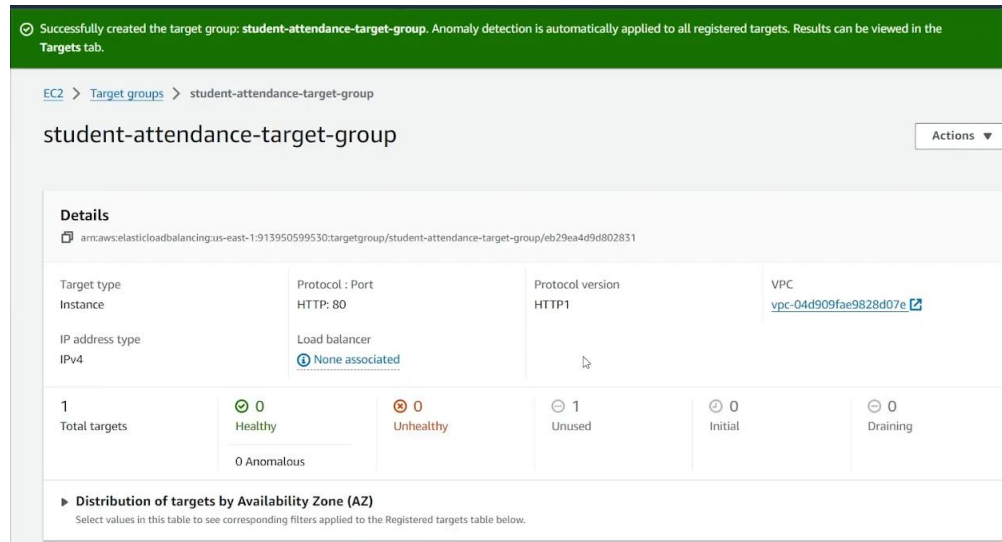
Targets (1)

Filter targets

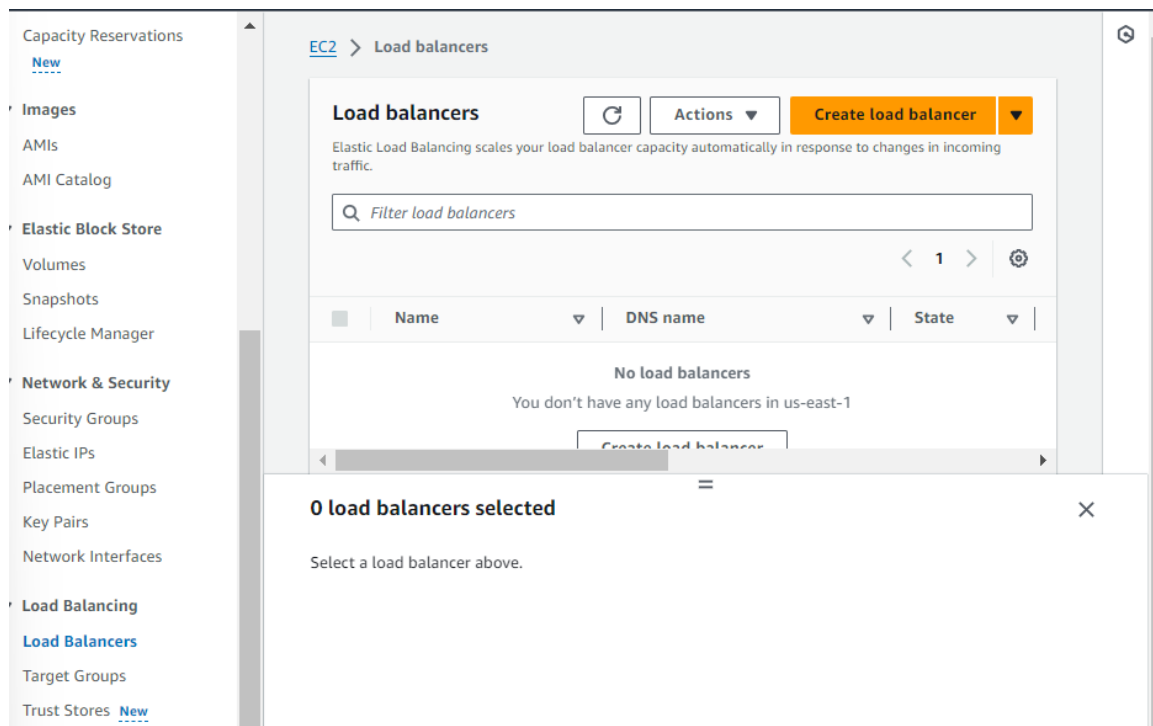
Show only pending

Remove all pending

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4
i-09dc6574d7842ee2b	student-attendance-server	80	Running	ec2-rds-1, server-security-group	us-east-1a	10.0.7.252



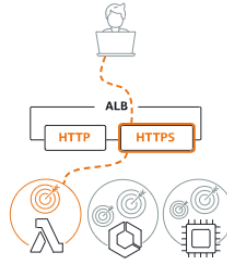
17. Then, scroll down to ‘Review targets’ and click “Create target group”. New target group successfully created.



18. Now, go to ‘Load Balancers’ on the left websites and click “Create load balancer” to create a load balancer.

Load balancer types

Application Load Balancer [Info](#)



Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

Create

19. Choose an application load balancer for the Load balancer types.

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

student-attendance-load-balancer

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)

Scheme can't be changed after the load balancer is created.

☒ Internet-facing

An internet-facing load balancer routes requests from clients over the internet to targets. Requires a public subnet. [Learn more](#)

☐ Internal

An internal load balancer routes requests from clients to targets using private IP addresses. Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type [Info](#)

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ IPv4

Includes only IPv4 addresses.

☐ Dualstack

Includes IPv4 and IPv6 addresses.

☐ Dualstack without public IPv4

Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with internet-facing load balancers only.

20. We used “student-attendance-load-balancer” as a load balancer name. Keep it default for ‘Scheme’ and ‘IP address type’.

Network mapping [Info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC [Info](#)

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#). For a new VPC, [create a VPC](#).

student-attendance-vpc
vpc-0746fdcc67a2457d0
IPv4 VPC CIDR: 10.0.0.0/16



21. Choose the same VPC that we created for the VPC. After choosing the VPC, we can find out the public subnet that we created before.

VPC [Info](#)

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#). For a new VPC, [create a VPC](#).

student-attendance-vpc
vpc-04d909fae9828d07e
IPv4 VPC CIDR: 10.0.0.0/16

Mappings [Info](#)

Select at least two Availability Zones and one subnet per zone. The load balancer routes traffic to targets in these Availability Zones only. Availability Zones that are not supported by the load balancer or the VPC are not available for selection.

Availability Zones

☒ **us-east-1a (use1-az1)**

Subnet

subnet-05be8676dc12b9f47 student-attendance-subnet-public1-us-east-1a
IPv4 subnet CIDR: 10.0.0.0/20

IPv4 address
Assigned by AWS

☒ **us-east-1b (use1-az2)**

Subnet

subnet-0f1e6fc140a057e79 student-attendance-subnet-public2-us-east-1b
IPv4 subnet CIDR: 10.0.16.0/20

IPv4 address
Assigned by AWS

☐ **us-east-1e (use1-az3)**

Subnet

subnet-00e92ad04d6aa053a student-public-1e
IPv4 subnet CIDR: 20.0.0.0/16

IPv4 address
Assigned by AWS

22. In mapping, we need to choose 3 availability zones which are us-east-1a, us-east-1b and us-east-1d and it must be public subnets.

Security groups [Info](#)

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups

Select up to 5 security groups



default
sg-0bddc374e64a804b9 VPC: vpc-0746fdcc67a2457d0

23. In the Security groups, remove the default security groups.

Security groups [Info](#)

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups

Select up to 5 security groups

ec2-rds-1
sg-00609d710c5295839 VPC: vpc-04d909fae9828d07e

server-security-group
sg-0c9f7a0340039d895 VPC: vpc-04d909fae9828d07e

24. Then, two security groups that we created before was added.

Listeners and routing [Info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80 Remove

Protocol: HTTP Port: 80

Default action: [Info](#)

Forward to: student-attendance-target-group HTTP

Target type: Instance, IPv4

[Create target group](#)

Listener tags - optional

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

Add listener tag

You can add up to 50 more tags.

25. In Listeners and routing, choose “HTTP” for protocol and select the target group that we created just now.

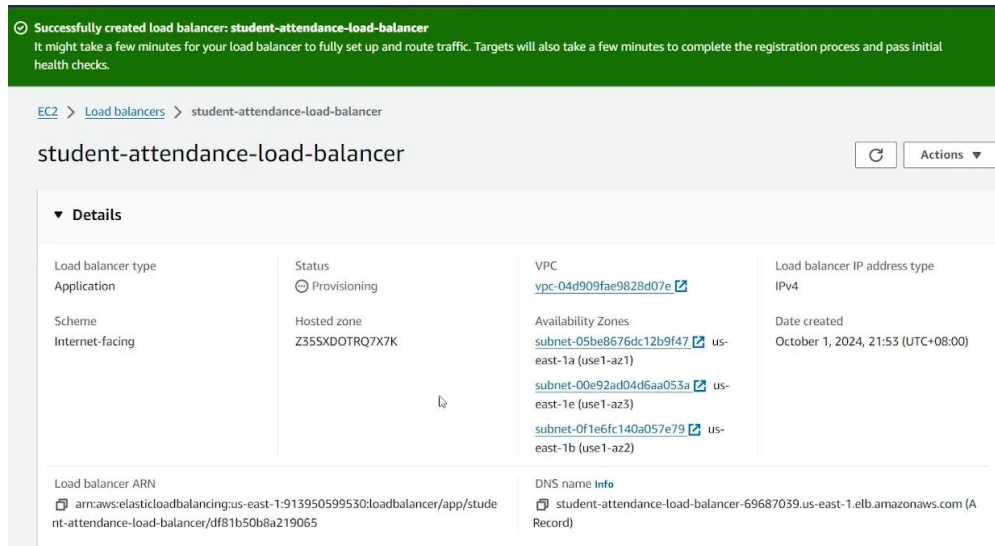
Review

Review the load balancer configurations and make changes if needed. After you finish reviewing the configurations, choose **Create load balancer**.

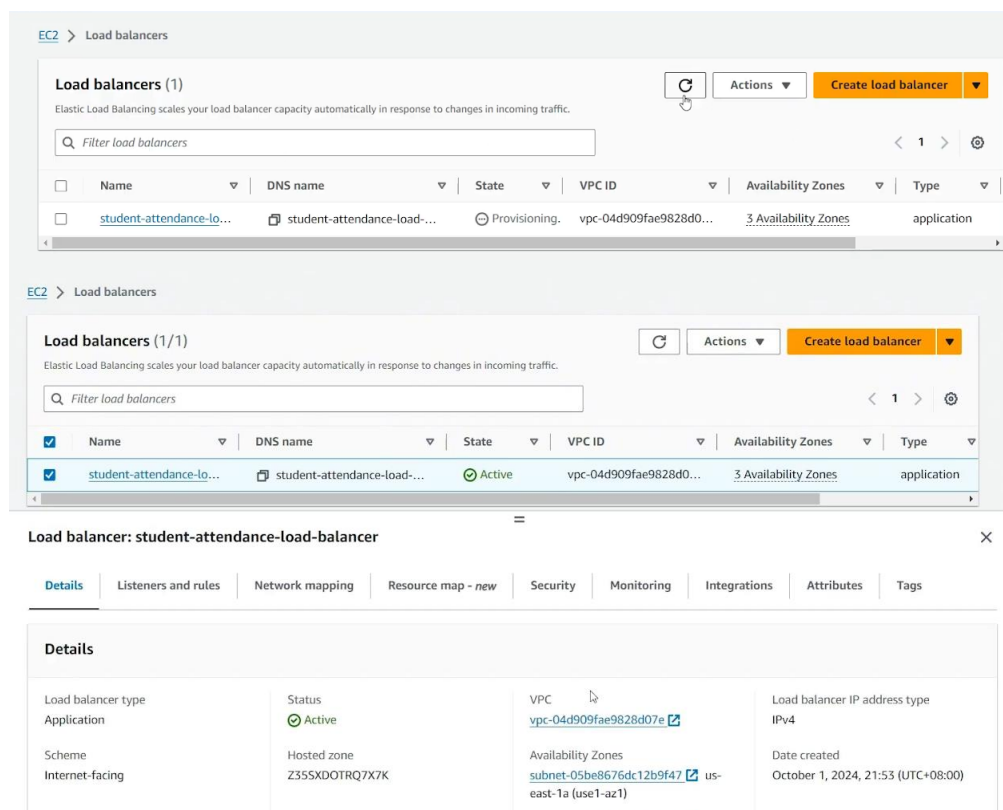
Summary

Review and confirm your configurations. [Estimate cost](#)

Basic configuration Edit	Security groups Edit	Network mapping Edit	Listeners and routing Edit
student-attendance-load-balancer - Internet-facing - IPv4	- ec2-rds-1 sg-00609d710c5295839 - server-security-group sg-0c9f7a0340039d895	VPC: vpc-04d909fae9828d07e student-attendance-vpc - us-east-1a subnet-05be8676dc12b9f47 student-attendance-subnet-public1-us-east-1a - us-east-1b subnet-0f1e6fc140a057e79 student-attendance-subnet-public2-us-east-1b - us-east-1e subnet-00e92ad04d6aa053a student-public-1e	HTTP:80 defaults to student-attendance-target-group
Service integrations Edit AWS WAF: None AWS Global Accelerator: None		Tags Edit None	



26. After finish setting up the configuration, click “Create load balance” and your load balancer was successfully created.

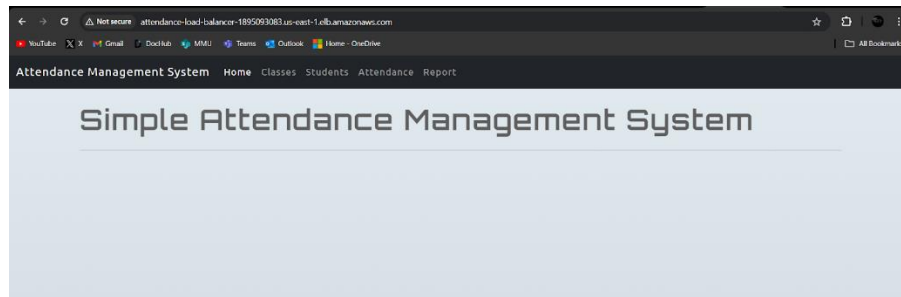


27. Return to Load balancers page, wait until the state change to “Active”.

ELB DNS to access the system.

Load balancer: student-attendance-load-balancer

Scheme Internet-facing	Hosted zone Z35SXDOTRQ7X7K	Availability Zones subnet-05be8676dc12b9f47 us-east-1a (use1-az1) subnet-00e92ad04d6aa053a us-east-1e (use1-az3) subnet-0f1e6fc140a057e79 us-east-1b (use1-az2)	Date created October 1, 2024, 21:53 (UTC+08:00)
Load balancer ARN arn:aws:elasticloadbalancing:us-east-1:913950599530:loadbalancer/app/student-attendance-load-balancer/df81b50b8a219065		DNS name copied student-attendance-load-balancer-69687039.us-east-1.elb.amazonaws.com (A Record)	



28. When the state changed to “Active”, copy the DNS name and paste it in the new tab. It will show our websites.

4.0 Phase 4

4.1 Implementation steps of Auto-Scaling setup

```
<?php
$servername = "studentattendancedatabase.cwoqz7r06eud.us-east-1.rds.amazonaws.com";
$username = "admin";
$password = "admin123";
$dbname = "simple_attendance_db";
$port = "3306";

$conn = new mysqli ($servername, $username, $password, $dbname);

if($conn->connect_error)
{
    die("Connection failed: " . $conn->connect_error);
}

?>
```

Upload succeeded
View details below.

The information below will no longer be available after you navigate away from this page.

Summary

Destination s3://student-att-bucket	Succeeded 1 file, 22.0 KB (100.00%)	Failed 0 files, 0 B (0%)
--	--	-----------------------------

Files and folders | Configuration

Files and folders (1 Total, 22.0 KB)

Find by name

Name	Folder	Type	Size	Status	Error
builtV22.zip	-	application/zip	22.0 KB	Succeeded	-

1. db-connect.php is needed to change the data connection details to make sure auto-scaling to function successfully before implementing auto scaling. Upload the new file to s3 bucket that we created in phase 1.

```

<?php

$servername = "studentattendancedatabase.cwoqz7r06eud.us-east-
1.rds.amazonaws.com";

$username = "admin";

$password = "admin123";

$dbname = "simple_attendance_db";

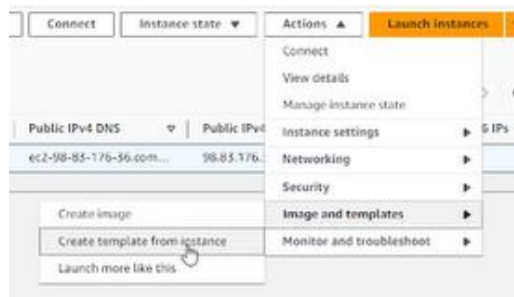
$port ="3306";

$conn = new mysqli ($servername, $username, $password, $dbname);

if($conn->connect_error)
{
die("Connection failed: " . $conn->connect_error);
}

?>

```



2. In the Instances page, select the instance. Go to ‘Actions’and ‘Image and templates’, click “Create template from instance”.

Launch template name and description

Source instance
i-09dc6574d7842ee2b

Launch template name - *required*
student-attendance-template
Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '*', '!', '(', ')', '[]'.

Template version description
version
Max 255 chars

Auto Scaling guidance [Info](#)
Select this if you intend to use this template with EC2 Auto Scaling
☒ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

Template tags

Summary

Software Image (AMI)
Amazon Linux 2 Kernel 5.10 AML...[read more](#)
ami-0e54eba7c51c234f6

Virtual server type (instance type)
t2.micro

Firewall (security group)
2 security groups

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 750 hours of public IPv4 address usage per month, 30 GiB of EBS storage, 2 million I/Os, 1 GB

Launch template contents

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

3. In the create launch template page, put “ student-attendance-template” as a launch template name and description put as a “version”. Make sure auto scaling guidance selected.

Network settings [Info](#)

Subnet [Info](#)
Don't include in launch template
When you specify a subnet, a network interface is automatically added to your template.

Firewall (security groups) [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Select existing security group ☐ Create security group

Common security groups [Info](#)
Select security groups
server-security-group sg-0c9f7a0340039d895 [X](#)
VPC: vpc-04d909fae9828d07e [Compare security group rules](#)

[Show all selected \(+1\)](#)
Security groups that you add or remove here will be added to or removed from all your network interfaces.

Advanced network configuration

Device index	Network interface	Description	
0	New interface		Remove

Existing network interfaces are not

2

Allow tags in metadata [Info](#)
Disable

User data - *optional* [Info](#)
Upload a file with your user data or enter it in the field.
[Choose file](#)

```
#!/bin/bash
sudo yum update -y
sudo amazon-linux-extras install -y php8.0 mariadb10.5
sudo yum install -y httpd
sudo systemctl start httpd
sudo systemctl enable httpd
sudo systemctl start MariaDB
sudo systemctl enable mariadb
sudo usermod -a -G apache ec2-user
sudo chown -R ec2-user:apache /var/www
sudo chmod 2775 /var/www
find /var/www -type d -exec sudo chmod 2775 {} \;
find /var/www -type f -exec sudo chmod 0664 {} \;
mkdir stud-att
cd stud-att
```

☐ User data has already been base64 encoded

Summary

Software Image (AMI)
Amazon Linux 2 Kernel 5.10 AML...[read more](#)
ami-0e54eba7c51c234f6

Virtual server type (instance type)
t2.micro

Firewall (security group)
2 security groups

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 750 hours of public IPv4 address usage per month, 30 GiB of EBS storage, 2 million I/Os, 1 GB

Cancel [Create launch template](#)

```
#!/bin/bash

sudo yum update -y

sudo amazon-linux-extras install -y php8.0 mariadb10.5

sudo yum install -y httpd

sudo systemctl start httpd

sudo systemctl enable httpd

sudo systemctl start MariaDB

sudo systemctl enable mariadb

sudo usermod -a -G apache ec2-user

sudo chown -R ec2-user:apache /var/www

sudo chmod 2775 /var/www

find /var/www -type d -exec sudo chmod 2775 {} \;

find /var/www -type f -exec sudo chmod 0664 {} \;

mkdir stud-att

cd stud-att

wget https://student-attendance-bucket.s3.amazonaws.com/builtV22.zip

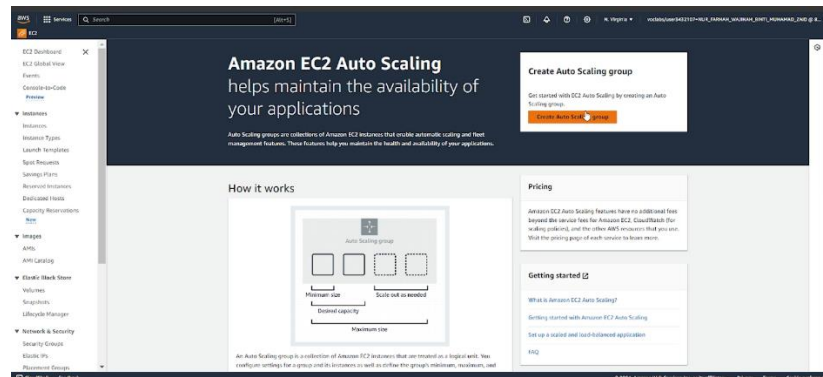
unzip builtV22.zip

cd built
```

🟢 **Success**
Successfully created student-attendance-template(tt-0b4ca8fa12ccfed9d).

▶ Actions log

4. In the network settings section, select “Don’t include in launch template” for the Subnet. Enter the same code that we used in phase 1 in the user data box. Then, create the template.



Choose launch template [info](#)

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.

Name

Auto Scaling group name

Enter a name to identify the group.

Must be unique to this account in the current Region and no more than 255 characters.

Choose a launch template that contains the instance-level settings, such as the Amazon Machine image (AMI), instance type, key pair, and security groups.

[Create a launch template](#)

Version

[Create a launch template version](#)

Description	Launch template	Instance type
version	student-attendance-template	t2.micro
AMI ID	lt-0b4ca8fa12ccfed9d	Request Spot Instances
ami-0e54eba7c51c234f6	Security groups	No
Key pair name	Security group IDs	
stud-att-keys	sg-00609d710c5295839	
	sg-0c9f7a0340039d895	

Additional details

Storage (volumes)	Date created
/dev/xvda	Tue Oct 01 2024 22:45:47 GMT+0800 (Malaysia Time)

5. Next, go to the auto-scaling group and click “Create Auto Scaling group”. In the choose launch template or configuration, put “student-attendance-auto-scaling-group” as an Auto Scaling group name and select the template that we created before.

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC
Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-04d909fae9828d07e (student-attendance-vpc) 10.0.0.0/16

[Create a VPC](#)

Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

us-east-1e | subnet-00e92ad04d6aa053a (student-public-1e) 20.0.0.0/16

us-east-1a | subnet-05be8676dc12b9f47 (student-attendance-subnet-public1-us-east-1a) 10.0.0.0/20

us-east-1b | subnet-0f1e6fc140a057e79 (student-attendance-subnet-public2-us-east-1b) 10.0.16.0/20

[Create a subnet](#)

Cancel Skip to review Previous **Next**

Configure advanced options - *optional* [Info](#)

Integrate your Auto Scaling group with other services to distribute network traffic across multiple servers using a load balancer or to establish service-to-service communications using VPC Lattice. You can also set options that give you more control over health check replacements and monitoring.

Load balancing [Info](#)

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

☐ No load balancer
Traffic to your Auto Scaling group will not be fronted by a load balancer.

☒ Attach to an existing load balancer
Choose from your existing load balancers.

☐ Attach to a new load balancer
Quickly create a basic load balancer to attach to your Auto Scaling group.

Attach to an existing load balancer
Select the load balancers that you want to attach to your Auto Scaling group.

☒ Choose from your load balancer target groups
This option allows you to attach Application, Network, or Gateway Load Balancers.

☐ Choose from Classic Load Balancers

Existing load balancer target groups
Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.

Select target groups

student-attendance-target-group | HTTP Application Load Balancer: student-attendance-load-balancer

6. In the network section, select VPC and availability zones and its subnets as shown in the figure above. Next, choose “Attach to an existing load balancer”, select “Choose from your load balancer target groups” and choose target groups that we created before in configure advanced options.

Health checks

Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.

EC2 health checks

[Always enabled](#)

Additional health check types - optional [Info](#)

☒ **Turn on Elastic Load Balancing health checks** **Recommended**

Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.

EC2 Auto Scaling will start to detect and act on health checks performed by Elastic Load Balancing. To avoid unexpected terminations, first verify the settings of these health checks in the [Load Balancer console](#).

☐ **Turn on VPC Lattice health checks**

VPC Lattice can monitor whether instances are available to handle requests. If it considers a target as failed a health check, EC2 Auto Scaling replaces it after its next periodic check.

☐ **Turn on Amazon EBS health checks**

EBS monitors whether an instance's root volume or attached volume stalls. When it reports an unhealthy volume, EC2 Auto Scaling can replace the instance on its next periodic health check.

Health check grace period [Info](#)

This time period delays the first health check until your instance finish initialization. It doesn't prevent an instance from termination when

Additional settings

Monitoring [Info](#)

☒ **Enable group metrics collection within CloudWatch**

Default instance warmup [Info](#)

The amount of time that CloudWatch metrics for new instances do not contribute to the group's aggregated instance metrics, as their usage data is not reliable yet.

☐ **Enable default instance warmup**

[Cancel](#)
[Skip to review](#)
[Previous](#)
[Next](#)

7. Then, tick on “ELB” for the health check type and in the Monitoring sections too. Then, click “Next”.

Configure group size and scaling - optional [Info](#)

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size [Info](#)

Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity

Specify your group size.

1

Scaling [Info](#)

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

1

Equal or less than desired capacity

Max desired capacity

3

Equal or greater than desired capacity

Automatic scaling - optional

Choose whether to use a target tracking policy [Info](#)

You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☒ No scaling policies

Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☐ Target tracking scaling policy

Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Instance maintenance policy [Info](#)

Control your Auto Scaling group's availability during instance replacement events. This includes health checks, instance refreshes, maximum instance lifetime features and events that happen automatically to keep your group balanced, called rebalancing events.

Choose a replacement behavior depending on your availability requirements

Mixed behavior

☒ No policy

For rebalancing events, new instances will launch before terminating others. For all other events, instances terminate and launch at the same time.

Prioritize availability

☐ Launch before terminating

Launch new instances and wait for them to be ready before terminating others. This allows you to go above your desired capacity by a given percentage and may temporarily increase costs.

Control costs

☐ Terminate and launch

Terminate and launch instances at the same time. This allows you to go below your desired capacity by a given percentage and may temporarily reduce availability.

Flexible

☐ Custom behavior

Set custom values for the minimum and maximum amount of available capacity. This gives you greater flexibility in setting how far below and over your desired capacity EC2 Auto Scaling goes when replacing instances.

Instance scale-in protection

Scale-in protection prevents newly launched instances from being terminated by scaling activities. Make sure to remove scale-in protection for the group or individual instances when instances are ready to be terminated.

☒ Enable instance scale-in protection

Cancel

Skip to review

Previous

Next

8. In configure group size and scaling, set maximum capacity to 3 to limit the instance number. Then, select “No scaling policies” for the Automatic scaling. Choose “No policy” for Instance maintenance policy. Before click “Next”, make sure instance scale-in protection is enabled.

Add notifications - optional [Info](#)

Send notifications to SNS topics whenever Amazon EC2 Auto Scaling launches or terminates the EC2 instances in your Auto Scaling group.

Add notification

Cancel

Skip to review

Previous

Next

Add tags - optional [Info](#)

Add tags to help you search, filter, and track your Auto Scaling group across AWS. You can also choose to automatically add these tags to instances when they are launched.

 You can optionally choose to add tags to instances (and their attached EBS volumes) by specifying tags in your launch template. We recommend caution, however, because the tag values for instances from your launch template will be overridden if there are any duplicate keys specified for the Auto Scaling group.

Tags (0)

Add tag

50 remaining

Cancel

Previous

Next

Instance maintenance policy

Replacement behavior

No policy

Min healthy percentage

-

Max healthy percentage

-

Instance scale-in protection

Instance scale-in protection

☒ Enable instance protection from scale in

Step 5: Add notifications

Edit

Notifications

No notifications

Step 6: Add tags

Edit

Tags (0)

Key	Value	Tag new instances
No tags		

Cancel

Previous

Create Auto Scaling group

9. Click “Skip to review” in the ‘Add notifications’ and ‘Add tags’. Scroll down and click “Create auto scaling group”.

EC2 > Auto Scaling groups

Auto Scaling groups (1) Info Refresh Launch configurations Launch templates Actions Create Auto Scaling group

Search your Auto Scaling groups

☐	Name	Launch template/configuration	Instances	Status	Desired capacity	Min	Max	Availabil...
☐	student-attendance-auto-scaling-group	student-attendance-template Version Dr	0	Updating capacity...	1	1	3	us-east-1a, ...

EC2 > Auto Scaling groups > student-attendance-auto-scaling-group

student-attendance-auto-scaling-group

Details Activity **Automatic scaling** Instance management Monitoring Instance refresh

Scaling policies resize your Auto Scaling group to meet changes in demand. With reactive dynamic scaling policies, you can track specific CloudWatch metrics and take action when the CloudWatch alarm threshold is met. Use predictive scaling policies along with dynamic scaling policies in the following situations: when your application demand changes quickly, but with a recurring pattern, or when your EC2 instances require more time to initialize.

Dynamic scaling policies (0) Info Refresh Actions Create dynamic scaling policy

No dynamic scaling policies have been created

Dynamic scaling policies use real-time data to scale your group based on configurable metrics.

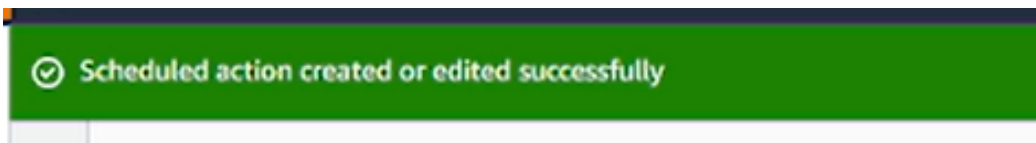
Create dynamic scaling policy

Scheduled actions (0) Info Refresh Actions Create scheduled action

Filter scheduled actions

☐	Name	Start time	End time	Recurrence	Time zone	Desired capacity	Min	Max
No scheduled actions are currently specified								

Create scheduled action



10. After creating a scaling group, go to automatic scaling to create scheduled action.

Instances (4) Info Last updated 1 minute ago Refresh Connect Instance state Actions Launch instances

Find Instance by attribute or tag (case-sensitive) All states

Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elas
Running	t2.micro	2/2 checks passed	View alarms	us-east-1e	ec2-54-198-69-236.co...	54.198.69.236	-
Running	t2.micro	2/2 checks passed	View alarms	us-east-1a	ec2-44-217-51-157.co...	44.217.51.157	44.2
Running	t2.micro	2/2 checks passed	View alarms	us-east-1a	ec2-35-175-209-170.co...	35.175.209.170	-
Running	t2.micro	2/2 checks passed	View alarms	us-east-1b	ec2-3-216-123-157.co...	3.216.123.157	-

Target group: student-attendance-target-group

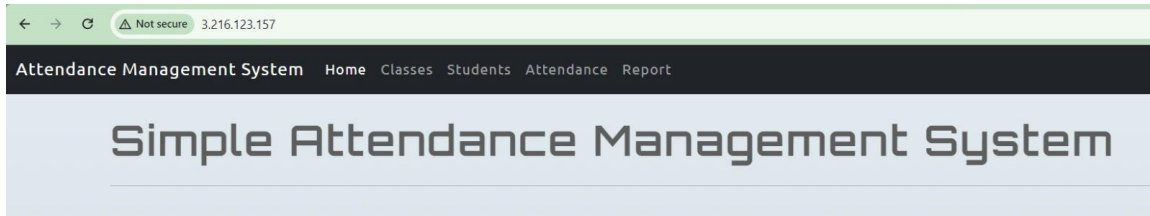
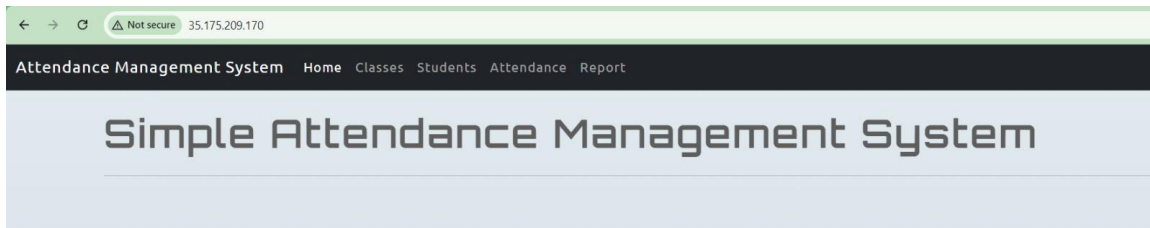
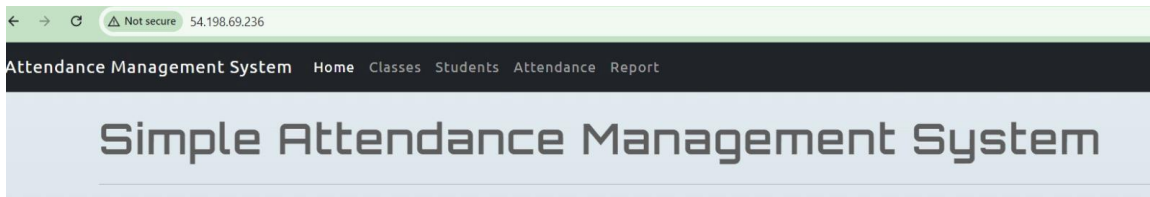


Registered targets (4) [Info](#) [Anomaly mitigation: Not applicable](#) [Deregister](#) [Register targets](#)

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

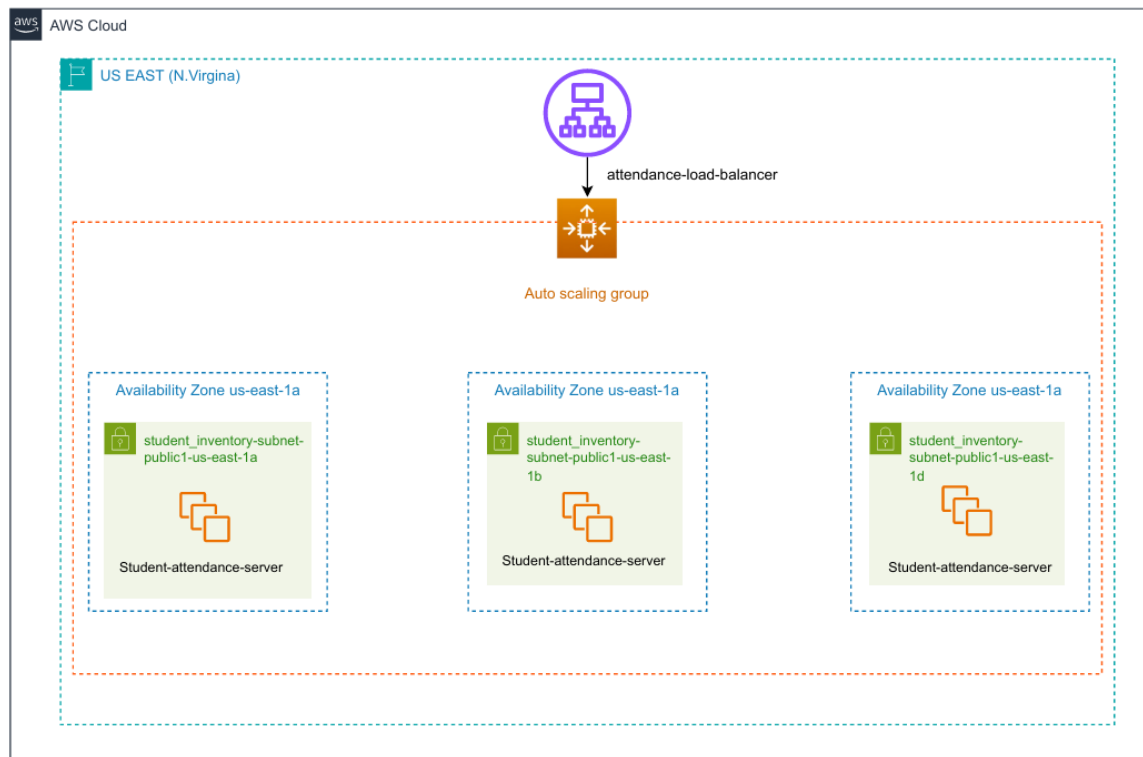
Filter targets

☐	Instance ID	Name	Port	Zone	Health status	Health status details	Launch...
☐	i-07e75765862da74af	student-atten...	80	us-east-1a	Healthy	-	October 3...
☐	i-0dfacbd56555016a3	student-atten...	80	us-east-1e	Healthy	-	October 2...
☐	i-02b901c4336a7cc18	student-atten...	80	us-east-1a	Healthy	-	October 2...
☐	i-09640eae2c97b5fef	student-atten...	80	us-east-1b	Healthy	-	October 2...



11. In a few minutes, there are 4 instances available which are 2 us-east-1a, 1 instance at us-east-1e, and 1 instance at us-east-1b availability zone. All the health status for each instance are healthy and public IP address can access to the website.

4.2 Layout diagram of the revised network



Reflection

During this project, we got a lot of experiences and lesson about AWS and self-development. Lessons and experiences we got during the making of this project are:

At first, we thought the time constraints in learner lab for 3 hours is not enough when we are trying to configure phase 1. However, when all phases are well-configured in sandbox, the estimation to do 1 phase is approximately 15 minutes each not including waiting time for creation such as instance and database creation.

In phase 1, the challenges we faced are the ssh connection of instance cannot be established which caused by the bug and personal laptop port is not open, the directory problem when deploying the website which caused by poor file location from existing file. Phase 1 requires basic linux command as the system will be hosted in Linux system.

In phase 2, the main problem every member faced is permission problem when connecting instance and database. The solution of this problem is by attaching LabRole to the IAM profile when accessing EC2. Instead of not connecting the database upon creation, we chose connecting to EC2 in pre-creation of the database in RDS.

In phase 3, there is no problem occur as it related to RDS. We learnt that RDS allowed creating the database right away in pre-creation of the system instead of creating the database from console.

Phase 4 is the most challenging part in this project because of the error only show up at the end after auto-scaling is implemented in the AWS system. The load balancer and the previous phase is well-configured, but the result of phase 4 that will allow the website to be accessed by 3 different IP addresses is not working well. However, after few days troubleshooting system in phase 4, we figured the problem is rooted from the mysql dump file that is not well configured before being uploaded and being processed by the launch template for autoscaling group.

Finally, this project not only helps us to understand AWS cloud system and configuration, but also teach us the teamwork among team members. Although one person has configured the system, every other member should also know how the system work for the long run. Every other member contributes to this project differently whether by offering the possible solutions, troubleshooting, getting information from various sources, and writing reports. Along the way, we have become a new person contained with AWS knowledge.

References

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 5. A Monk in Cloud. (2022, August 25). Host your personal site on AWS EC2 Instance | EC2 Instance | Demo | Website Hosting | Web App on EC2 [Video]. YouTube. <https://www.youtube.com/watch?v=Islmm-LMu38>
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Project Rubric

TCC6313 Cloud Computing (Trimester July/August 2420)

Project Rubric

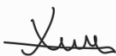
Evaluation Item	Demonstrated	Points Earned
Phase 1 (10 marks)		
- Functioning E-Commerce Asset Website (5 marks) - Homepage - View items - Add items - Edit items - Delete items		
- Network Layout (3 marks) - VPC - Gateway - Subnet		
- Web Application (2 marks) - EC2 - Accessible IP		
Phase 2 (5 marks)		
- A new private subnet (1 mark) - A new RDS MySQL instance (3marks) - Dump local MySQL (1 mark)		
Phase 3 (5 marks)		
- A new public subnet (1 mark) - Multi-AZ capability (3 marks) - Accessible ELB DNS (1 mark)		
Phase 4 (5 marks)		
- Auto Scaling properties: - EC2 (1 mark) - Number of instances (2 marks) - Scaling policies (2 marks)		
Total Points		/ 25

Report (10 marks)	Excellent	Good	Adequate	Marks
Implementation	4	3-2	1-0	
	Demonstrate complete steps of design implementation	Demonstrate partial steps of design implementation	Demonstrate very little steps of design implementation	
Network Layout	4	3-2	1-0	
	Provide very clear and concise layout of the design.	Provide a clear and inaccurate layout of the design.	Provide somewhat unclear and false layout of the design.	
Report Completeness	2	1	0	
	The report is comprehensive and complete	The report is somewhat complete	The report is incomplete	
Presentation (5 marks)	Excellent	Good	Adequate	Marks
Knowledge	5-4	3-2	1-0	
	Able to demonstrate a complete implementation of design requirements	Able to demonstrate the partial implementation of design requirements	Able to demonstrate very little implementation of design requirements	
Total Points				/15

Remarks:

DECLARATION: We agree that all members deserve equal marks from this assignment.

Write down the Stud ID (clearly) and signed (by all members):

Student ID	Student Name	Signature
1211103698	UMMI SYAHIRAH BINTI MUHAMMAD ROZAIDEE @ ROZI	
1211110723	U BOON XUAN	
1221301543	NUR FARHAH WAJIAH BINTI MUHAMAD ZAID	

1221302732	HANI HAMIZAH BINTI HAMZAN	
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